TERRAFORM - Knowledge Base JSON Files

Contents of every JSON file in the project.

1. kb/kb.json
2. kb/all_cases.json
3. kb/case1_arid.json
4. kb/case2_wetland.json
5. kb/case3_saline.json
6. kb/case4_pasture.json
7. kb/case5_deforested.json
8. knowledge_base.json

1. kb/kb.json

Full file contents


```
{
 "meta": {
  "name": "TERRAFORM KB - case3_saline",
  "kb_version": "0.1.0-draft",
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    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
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   "CONDITIONALLY_READY",
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    "F-RAIN-RARE"
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   "type": "internal"
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   "title": "TERRAFORM PRD",
   "type": "internal"
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   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
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   "id": "F-GRITTY",
   "label": "Gritty, loose, sandy texture",
   "kind": "text",
   "lexicon": "grit|sandy|\\bsand\\b|beach"
  },
  {
   "id": "F-SALT-CRUST",
   "label": "White or pale crust on surface",
   "kind": "visual",
   "lexicon": "white.{0,25}(crust|residue)|salt crust|crusty|crusted"
  },
  {
   "id": "F-SALT-SMELL",
   "label": "Salty air or smell",
   "kind": "text",
   "lexicon": "salt(y)? (smell|air|wind|spray)|smell of salt|salty"
  },
  {
   "id": "F-STUNTED-BROWN",
   "label": "Stunted, browning vegetation",
   "kind": "visual",
   "lexicon": "stunted|browning|scorched"
  },
  {
   "id": "F-SEA-NEAR",
   "label": "Sea or shoreline nearby",
   "kind": "text",
   "lexicon": "\\bsea\\b|shore|coast|ocean"
  }
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   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
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   "id": "T-SLOPE",
   "label": "Sloped Terrain",
   "sets": {
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   }
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   "label": "Rocky",
   "sets": {
    "rocky": true
   }
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   "id": "T-SHADE",
   "label": "Shaded",
   "sets": {
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   "cancels_features": [
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   "label": "Recent Rain",
   "sets": {
    "recent_rain": true
   }
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   "id": "T-WIND",
   "label": "Strong Wind",
   "sets": {
    "strong_wind": true
   }
  },
  {
   "id": "T-SALT-AIR",
   "label": "Salty/Coastal Air",
   "sets": {
    "salt_air": true
   },
   "adds_features": [
    "F-SALT-SMELL"
   ]
  }
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 "archetypes": [
  {
   "id": "ARC-SANDY",
   "name": "Sandy / Loamy Sand (Degraded)",
   "criteria": "Gritty, loose texture; low cohesion; rapid drainage.",
   "physical_profile": "Loose, particulate, low aggregate stability",
   "core_gaps": [
    "low water-holding capacity",
    "low organic matter",
    "low nutrient retention"
   ],
   "typical_estimates": {
    "texture_class": "Sand / loamy sand",
    "drainage": "Excessive",
    "ph_range": "5.5-7.0",
    "organic_matter": "Low",
    "cec": "Low"
   },
   "estimate_label": "estimate, not laboratory measurement",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "stress_tags": [
    "drought",
    "low fertility"
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  },
  {
   "id": "ARC-SALINE",
   "name": "Saline / Salt-Affected",
   "criteria": "Salt crust or salty air, stunted browning plants, coastal context.",
   "variants": [
    "visible crust",
    "no crust but salty smell and stunting",
    "salt spray without saline soil"
   ],
   "core_gaps": [
    "salinity",
    "wind exposure",
    "low water retention if sandy"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "stress_tags": [
```

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    "salinity"
   ],
   "typical_estimates": {
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    "drainage": "Rapid",
    "organic_matter": "Low",
    "cec": "Low",
    "ph_range": "7.5-8.5"
   }
  }
 ],
 "rules": [
  {
   "id": "R-SANDY-1",
   "type": "diagnostic",
   "archetype": "ARC-SANDY",
   "when": {
    "all": [
     "F-GRITTY"
    ]
   },
   "concludes": "Sandy texture. Texture is not by itself degradation; confirm low organic matter before diagnosing
     degradation.",
   "min_evidence": 1,
   "status": "pending_review"
  },
  {
   "id": "R-SALINE-1",
   "type": "diagnostic",
   "archetype": "ARC-SALINE",
   "when": {
    "any": [
     "F-SALT-CRUST",
     "F-SALT-SMELL"
    ],
    "also_any": [
     "F-STUNTED-BROWN",
     "F-SEA-NEAR"
    ]
   },
   "concludes": "Probable salinity. Salinity can only be confirmed by an EC measurement.",
   "min_evidence": 2,
   "status": "pending_review"
  },
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   "id": "R-WATER-SAFETY",
   "legacy_name": "R14",
   "type": "safety_filter",
   "when": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect_data": {
    "require_water": [
     "low",
     "low-medium"
    ]
   }
  },
  {
   "id": "R-HEAVY-FEEDER",
   "legacy_name": "R13",
   "type": "safety_filter",
   "when": {
    "deficiency_any": [
     "low organic matter",
     "nutrient depletion",
     "low nutrient retention"
    ]
   },
   "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "archetype_stress_any": [
     "low fertility"
    ]
   },
   "effect_data": {
    "exclude_feeder_class": [
     "heavy"
    ]
```

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   }
  },
  {
   "id": "R-READINESS-GATE",
   "legacy_name": "R9",
   "type": "readiness",
   "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
     counts as not passed.",
   "status": "pending_review"
  },
  {
   "id": "R-MIXED",
   "type": "combination",
   "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
   "status": "pending_review"
  },
  {
   "id": "R-NO-ESTIMATE-AS-FACT",
   "type": "integrity",
   "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
     labelled 'measured'.",
   "status": "pending_review"
  }
 ],
 "procedures": [
  {
   "id": "PROC-ORG-MATTER",
   "name": "Compost / well-rotted manure incorporation",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "targets": [
    "low organic matter",
    "low water retention",
    "nutrient depletion"
   ],
   "prerequisites": [],
   "materials": [
    "compost or well-rotted manure"
   ],
   "rate": 3,
   "unit": "kg per m2",
   "steps": [
    "Spread evenly over the plot.",
    "Incorporate into the topsoil.",
    "Water lightly if water is available."
   ],
   "verification": [
    "Crumb structure test"
   ],
   "stabilization_days": 7,
   "safety": [
    "Use fully rotted material only."
   ],
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-MULCH",
   "name": "Organic mulch cover",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-SALINE",
    "ARC-DISTURBED"
   ],
   "targets": [
    "moisture loss",
    "surface protection",
    "wind erosion"
   ],
   "prerequisites": [],
   "materials": [
    "straw, dry grass or crop residue"
   ],
   "rate": 5,
   "unit": "cm depth",
   "steps": [
    "Apply after amendments are incorporated.",
    "Keep mulch in place against wind (anchor or wet down)."
   ],
   "verification": [
    "Soil under mulch stays visibly moist"
   ],
   "stabilization_days": 0,
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
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 {
  "id": "PROC-BIOCHAR",
  "name": "Biochar incorporation",
  "applies_to": [
   "ARC-SANDY"
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  "targets": [
   "low nutrient retention",
   "low water retention"
  ],
  "prerequisites": [
   "PROC-ORG-MATTER"
  ],
  "materials": [
   "biochar, pre-charged with compost"
  ],
  "rate": 0.5,
  "unit": "kg per m2",
  "steps": [
   "Mix with compost before applying."
  ],
  "verification": [],
  "stabilization_days": 7,
  "phase": "B",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
 {
  "id": "PROC-WINDBREAK",
  "name": "Windbreak and ground cover",
  "applies_to": [
   "ARC-ARID",
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "targets": [
   "wind erosion",
   "wind damage"
  ],
  "prerequisites": [],
  "materials": [
   "tall hardy plants, brush or woven barriers"
  ],
  "rate": "1 barrier row per 10 m of exposed plot width, across the prevailing wind",
  "unit": "",
  "steps": [
   "Place barriers across the prevailing wind direction.",
   "Cover bare patches before planting."
  ],
  "verification": [
   "Visible dust movement reduced"
  ],
  "stabilization_days": 0,
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-CASE"
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 {
  "id": "PROC-LEACH",
  "name": "Salinity leaching",
  "applies_to": [
   "ARC-SALINE"
  ],
  "targets": [
   "salinity"
  ],
  "prerequisites": [
   "Confirmed low-salt water source",
   "Confirmed drainage out of the root zone"
  ],
  "materials": [
   "low-salt water"
  ],
  "rate": 150,
  "unit": "mm of low-salt water per leaching event; repeat until crust is gone",
  "steps": [
   "Flush with low-salt water only if water and drainage are confirmed."
  ],
  "verification": [
   "EC measurement or crust gone"
  ],
  "stabilization_days": 7,
  "safety": [
   "Leaching without drainage raises the water table and brings salt back up."
  ],
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
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],
"readiness_criteria": [
 {
  "id": "RC-CRUMB",
```

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   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-MOIST",
   "name": "Moisture retention",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID"
   ],
   "check": "Soil stays damp below the surface after watering.",
   "mandatory": true,
   "if_failed": "Add organic matter and mulch."
  },
  {
   "id": "RC-DUST",
   "name": "Surface stability",
   "applies_to": [
    "ARC-ARID",
    "ARC-SANDY",
    "ARC-SALINE"
   ],
   "check": "Wind no longer lifts loose soil from the surface.",
   "mandatory": true,
   "if_failed": "Add windbreak or ground cover."
  },
  {
   "id": "RC-SALT",
   "name": "Salt crust cleared",
   "applies_to": [
    "ARC-SALINE"
   ],
   "check": "No visible crust and a measured EC reading is available.",
   "mandatory": true,
   "if_failed": "Provide an EC reading or continue salinity management."
  },
  {
   "id": "RC-STAB",
   "name": "Stabilization window",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Required waiting time after amendments has passed.",
   "mandatory": true,
   "window_days": 3,
   "if_failed": "Wait for the stabilization window to pass, then repeat."
  }
 ],
 "crops": [
  {
   "id": "CROP-PMILLET",
   "name": "Pearl millet",
   "scientific": "Pennisetum glaucum",
   "category": "Cereal / dryland grain",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-SALINE",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility",
    "some salinity"
   ],
   "avoid": [
    "waterlogging"
   ],
   "water": "low",
   "light": "full",
   "duration_days": [
    65,
    90
   ],
   "feeder_class": "light",
   "role": "Drought and low-fertility tolerant cereal.",
   "planting": {
    "depth": "2-3 cm",
    "spacing": "10-15 cm in row, 45-50 cm between rows",
```

```
   "method": "Direct sow into prepared soil"
  },
  "care": {
   "diseases": [
    "Downy mildew",
    "Ergot"
   ],
   "pests": [
    "Stem borer"
   ],
   "controls": [
    "Clean or resistant seed",
    "Remove infected plants early",
    "Neem spray"
   ],
   "fertilizer": "Low demand; compost at bed prep and a small nitrogen top-dress at tillering."
  },
  "harvest": {
   "indicators": [
    "Ear heads dry, grains hard"
   ],
   "method": "Cut ear heads and dry"
  },
  "status": "pending_review",
  "source": "SRC-PRD",
  "phase_days": {
   "P1": "3-5",
   "P2": "20-25",
   "P3": "25-35",
   "P4": "7-10"
  }
 },
 {
  "id": "CROP-BARLEY",
  "name": "Barley",
  "scientific": "Hordeum vulgare",
  "category": "Cereal",
  "soil_compat": [
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "tolerates": [
   "salinity",
   "drought"
  ],
  "avoid": [
   "waterlogging"
  ],
  "water": "low-medium",
  "light": "full",
  "duration_days": [
   90,
   120
  ],
  "feeder_class": "moderate",
  "role": "Salt-tolerant cereal.",
  "planting": {
   "depth": "3-5 cm",
   "spacing": "Drill in rows 20 cm apart",
   "method": "Drill or broadcast and cover"
  },
  "care": {
   "diseases": [
    "Powdery mildew",
    "Net blotch",
    "Rust"
   ],
   "pests": [
    "Aphids",
    "Armyworm"
   ],
   "controls": [
    "Resistant varieties",
    "Crop rotation",
    "Neem spray"
   ],
   "fertilizer": "Moderate nitrogen, split; compost at bed prep; avoid chloride-heavy fertilizers."
  },
  "harvest": {
   "indicators": [
    "Heads golden",
    "Grains hard"
   ],
   "method": "Cut when dry, thresh, store below 12% moisture"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
   "P1": "5-10",
   "P2": "30-40",
   "P3": "30-40",
   "P4": "10-14"
  }
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```
  },
  {
   "id": "CROP-TOMATO",
   "name": "Tomato",
   "scientific": "Solanum lycopersicum",
   "category": "Vegetable",
   "soil_compat": [],
   "tolerates": [],
   "avoid": [
    "drought",
    "low fertility",
    "salinity"
   ],
   "water": "high",
   "light": "full",
   "duration_days": [
    90,
    120
   ],
   "feeder_class": "heavy",
   "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
   "planting": {
    "depth": "0.5-1 cm in nursery; transplant at 5 cm",
    "spacing": "45 x 60 cm",
    "method": "Raise seedlings, then transplant"
   },
   "care": {
    "diseases": [
     "Early blight",
     "Late blight",
     "Bacterial wilt"
    ],
    "pests": [
     "Whitefly",
     "Fruit borer"
    ],
    "controls": [
     "Neem spray",
     "Crop rotation",
     "Stake and prune for airflow"
    ],
    "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
   },
   "harvest": {
    "indicators": [
     "Fruit fully coloured and slightly firm"
    ],
    "method": "Hand-pick ripe fruit regularly"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
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    "P2": "25-30",
    "P3": "30-45",
    "P4": "7-14"
   }
  }
 ],
 "phases": {
  "journey": [
   {
    "id": "P1",
    "name": "Germination"
   },
   {
    "id": "P2",
    "name": "Vegetative Growth"
   },
   {
    "id": "P3",
    "name": "Reproductive & Canopy Development"
   },
   {
    "id": "P4",
    "name": "Pre-Harvest"
   }
  ],
  "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
     data is unavailable."
 },
 "missing_data_messages": {
  "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
  "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
  "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
  "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
  "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
 }
}
```

2. kb/all_cases.json

Full file contents


```
{
 "meta": {
  "name": "TERRAFORM KB - all cases",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ],
  "water_confirm_features": [
   "F-STANDING-WATER"
  ],
  "conflicts": [
   [
    "F-STANDING-WATER",
    "F-CRACKED"
   ],
   [
    "F-STANDING-WATER",
    "F-RAIN-RARE"
   ]
  ],
  "text_limit": 3000
 },
 "sources": [
  {
   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
  },
  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
  },
  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
 ],
 "features": [
  {
   "id": "F-CRACKED",
   "label": "Cracked soil surface",
   "kind": "visual",
   "lexicon": "crack|plates|fissur"
  },
  {
   "id": "F-LOOSE-DUST",
   "label": "Loose soil or dust lifted by wind",
   "kind": "visual",
   "lexicon": "dust|wind[- ]?(blown|lifting)|dust|lifting loose|loose soil|erosion|eroding"
  },
  {
   "id": "F-GRITTY",
   "label": "Gritty, loose, sandy texture",
   "kind": "text",
   "lexicon": "grit|sandy|\\bsand\\b|beach"
  },
  {
   "id": "F-NO-VEG",
   "label": "No or very sparse vegetation",
   "kind": "visual",
   "lexicon": "no (visible )?(veg|plants|green)|barren|bare (land|plot|ground)|sparse|scrap of green|dead
     (veg|weeds)|brittle weeds"
  },
  {
   "id": "F-STANDING-WATER",
   "label": "Standing water or puddles",
   "kind": "visual",
   "lexicon": "standing (water|puddle)|puddle|pooling|submerged|flooded"
  },
  {
   "id": "F-SOUR-SMELL",
   "label": "Sour or musty soil smell",
   "kind": "text",
   "lexicon": "sour|musty|foul smell"
  },
  {
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  "id": "F-YELLOW-WEEDS",
  "label": "Yellowing weeds with soft or rotting roots",
  "kind": "visual",
  "lexicon": "yellow|rotting roots|soft roots|wilting"
 },
 {
  "id": "F-SLOW-DRAIN",
  "label": "Drains slowly, stays wet after rain",
  "kind": "text",
  "lexicon": "slow(ly)? drain|drains slowly|stays wet|waterlog|poorly drain"
 },
 {
  "id": "F-DARK-STICKY",
  "label": "Dark, sticky clay",
  "kind": "text",
  "lexicon": "sticky|dark clay|heavy clay"
 },
 {
  "id": "F-SALT-CRUST",
  "label": "White or pale crust on surface",
  "kind": "visual",
  "lexicon": "white.{0,25}(crust|residue)|salt crust|crusty|crusted"
 },
 {
  "id": "F-SALT-SMELL",
  "label": "Salty air or smell",
  "kind": "text",
  "lexicon": "salt(y)? (smell|air|wind|spray)|smell of salt|salty"
 },
 {
  "id": "F-STUNTED-BROWN",
  "label": "Stunted, browning vegetation",
  "kind": "visual",
  "lexicon": "stunted|browning|scorched"
 },
 {
  "id": "F-SEA-NEAR",
  "label": "Sea or shoreline nearby",
  "kind": "text",
  "lexicon": "\\bsea\\b|shore|coast|ocean"
 },
 {
  "id": "F-HARD-PACKED",
  "label": "Hard-packed ground",
  "kind": "text",
  "lexicon": "hard[- ]packed|compact|hard ground|pressed[- ]down|rock[- ]hard"
 },
 {
  "id": "F-SURFACE-POOLING",
  "label": "Water pools on surface instead of soaking in",
  "kind": "text",
  "lexicon": "water (sits|pools)|pools on (the )?surface|doesn.t soak|not soak|poor infiltration"
 },
 {
  "id": "F-BARE-PATCHES",
  "label": "Bare patches",
  "kind": "visual",
  "lexicon": "bare (patch|dirt|spot)|worn (down )?to"
 },
 {
  "id": "F-THIN-GRASS",
  "label": "Thin, patchy, grazed grass",
  "kind": "visual",
  "lexicon": "thin|patchy|overgraz|grazed|grazing|pasture"
 },
 {
  "id": "F-STUMPS",
  "label": "Stumps, debris or vehicle tracks",
  "kind": "visual",
  "lexicon": "stump|debris|tracks|cleared|deforest|felled|logging"
 },
 {
  "id": "F-NO-CANOPY",
  "label": "No canopy or shade",
  "kind": "visual",
  "lexicon": "no (canopy|shade|trees)|exposed to (the )?sun|full sun|open to the sun|cleared"
 },
 {
  "id": "F-HIGH-HEAT",
  "label": "Daytime temperatures reported above 40 C",
  "kind": "text",
  "lexicon": "\\b4\\d ?\u00b0? ?c|scorching|extreme heat|very hot|intense heat"
 },
 {
  "id": "F-RAIN-RARE",
  "label": "Rainfall rare or unreliable",
  "kind": "text",
  "lexicon": "rain(fall)? (is )?(rare|unreliable|unpredictable|scarce)|rare.{0,30}rain|no rain|last rain"
 }
],
"toggles": [
 {
  "id": "T-WATER",
```

```
   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
  },
  {
   "id": "T-SLOPE",
   "label": "Sloped Terrain",
   "sets": {
    "slope": true
   }
  },
  {
   "id": "T-ROCKY",
   "label": "Rocky",
   "sets": {
    "rocky": true
   }
  },
  {
   "id": "T-SHADE",
   "label": "Shaded",
   "sets": {
    "shaded": true
   },
   "cancels_features": [
    "F-NO-CANOPY"
   ]
  },
  {
   "id": "T-RAIN",
   "label": "Recent Rain",
   "sets": {
    "recent_rain": true
   }
  },
  {
   "id": "T-WIND",
   "label": "Strong Wind",
   "sets": {
    "strong_wind": true
   }
  },
  {
   "id": "T-SALT-AIR",
   "label": "Salty/Coastal Air",
   "sets": {
    "salt_air": true
   },
   "adds_features": [
    "F-SALT-SMELL"
   ]
  }
 ],
 "archetypes": [
  {
   "id": "ARC-SANDY",
   "name": "Sandy / Loamy Sand (Degraded)",
   "criteria": "Gritty, loose texture; low cohesion; rapid drainage.",
   "physical_profile": "Loose, particulate, low aggregate stability",
   "core_gaps": [
    "low water-holding capacity",
    "low organic matter",
    "low nutrient retention"
   ],
   "typical_estimates": {
    "texture_class": "Sand / loamy sand",
    "drainage": "Excessive",
    "ph_range": "5.5-7.0",
    "organic_matter": "Low",
    "cec": "Low"
   },
   "estimate_label": "estimate, not laboratory measurement",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "stress_tags": [
    "drought",
    "low fertility"
   ]
  },
  {
   "id": "ARC-ARID",
   "name": "Arid / Drought-Stressed",
   "criteria": "Dry, low organic matter, poor moisture retention, unreliable rain.",
   "variants": [
    "cracked-compacted (retention dominant)",
    "loose-eroding (wind erosion dominant)"
   ],
   "core_gaps": [
    "water scarcity",
    "wind erosion",
    "poor structure"
   ],
```

```
 "status": "pending_review",
 "source": "SRC-CASE",
 "stress_tags": [
  "drought",
  "heat",
  "low fertility"
 ],
 "typical_estimates": {
  "texture_class": "Clay-loam crust to loose sand",
  "drainage": "Poor when crusted, rapid when sandy",
  "organic_matter": "Low",
  "cec": "Low to moderate",
  "ph_range": "7.0-8.5"
 }
},
{
 "id": "ARC-WATERLOGGED",
 "name": "Waterlogged / Poorly Drained",
 "criteria": "Standing water or slow drainage; sour smell; oxygen-starved root zone.",
 "variants": [
  "recently flooded (days)",
  "prolonged waterlogging",
  "heavy slow-draining without visible pooling"
 ],
 "core_gaps": [
  "drainage",
  "root-zone oxygen",
  "nutrient loss",
  "disease risk"
 ],
 "status": "pending_review",
 "source": "SRC-CASE",
 "stress_tags": [
  "waterlogging"
 ],
 "typical_estimates": {
  "texture_class": "Clay / silty clay",
  "drainage": "Very poor",
  "organic_matter": "Moderate, partly decomposed",
  "cec": "Moderate to high",
  "ph_range": "5.5-6.5"
 }
},
{
 "id": "ARC-SALINE",
 "name": "Saline / Salt-Affected",
 "criteria": "Salt crust or salty air, stunted browning plants, coastal context.",
 "variants": [
  "visible crust",
  "no crust but salty smell and stunting",
  "salt spray without saline soil"
 ],
 "core_gaps": [
  "salinity",
  "wind exposure",
  "low water retention if sandy"
 ],
 "status": "pending_review",
 "source": "SRC-CASE",
 "stress_tags": [
  "salinity"
 ],
 "typical_estimates": {
  "texture_class": "Sand / loamy sand",
  "drainage": "Rapid",
  "organic_matter": "Low",
  "cec": "Low",
  "ph_range": "7.5-8.5"
 }
},
{
 "id": "ARC-COMPACTED",
 "name": "Compacted and Depleted",
 "criteria": "Hard-packed ground, poor infiltration, bare or thin grazed vegetation, low organic matter.",
 "variants": [
  "uniform",
  "uneven wear"
 ],
 "core_gaps": [
  "compaction",
  "poor infiltration",
  "nutrient and organic matter depletion"
 ],
 "status": "pending_review",
 "source": "SRC-CASE",
 "stress_tags": [
  "low fertility"
 ],
 "typical_estimates": {
  "texture_class": "Compacted loam / clay loam",
  "drainage": "Poor infiltration",
  "organic_matter": "Low",
  "cec": "Moderate",
```

```
    "ph_range": "5.5-7.0"
   }
  },
  {
   "id": "ARC-DISTURBED",
   "name": "Disturbed, Compacted, Sun-Exposed",
   "criteria": "Cleared land with tracks, stumps or debris; no canopy; compacted.",
   "core_gaps": [
    "compaction",
    "heat and sun exposure",
    "no surface cover"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "stress_tags": [
    "heat",
    "low fertility"
   ],
   "typical_estimates": {
    "texture_class": "Compacted loam / clay loam",
    "drainage": "Poor infiltration",
    "organic_matter": "Low (topsoil lost)",
    "cec": "Low to moderate",
    "ph_range": "5.5-7.0"
   }
  }
 ],
 "rules": [
  {
   "id": "R-ARID-1",
   "type": "diagnostic",
   "archetype": "ARC-ARID",
   "when": {
    "all": [
     "F-NO-VEG"
    ],
    "any": [
     "F-CRACKED",
     "F-RAIN-RARE",
     "F-HIGH-HEAT"
    ]
   },
   "concludes": "Drought stress with poor moisture retention",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-ARID-2",
   "type": "diagnostic",
   "archetype": "ARC-ARID",
   "when": {
    "all": [
     "F-LOOSE-DUST"
    ],
    "any": [
     "F-NO-VEG",
     "F-GRITTY",
     "F-CRACKED"
    ]
   },
   "concludes": "Active wind erosion in addition to dryness",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-SANDY-1",
   "type": "diagnostic",
   "archetype": "ARC-SANDY",
   "when": {
    "all": [
     "F-GRITTY"
    ]
   },
   "concludes": "Sandy texture. Texture is not by itself degradation; confirm low organic matter before diagnosing
     degradation.",
   "min_evidence": 1,
   "status": "pending_review"
  },
  {
   "id": "R-WET-1",
   "type": "diagnostic",
   "archetype": "ARC-WATERLOGGED",
   "when": {
    "any": [
     "F-STANDING-WATER",
     "F-SLOW-DRAIN"
    ],
    "also_any": [
     "F-SOUR-SMELL",
     "F-YELLOW-WEEDS",
     "F-DARK-STICKY"
    ]
   },
```

```
 "concludes": "Waterlogging with root-zone oxygen stress",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-SALINE-1",
 "type": "diagnostic",
 "archetype": "ARC-SALINE",
 "when": {
  "any": [
   "F-SALT-CRUST",
   "F-SALT-SMELL"
  ],
  "also_any": [
   "F-STUNTED-BROWN",
   "F-SEA-NEAR"
  ]
 },
 "concludes": "Probable salinity. Salinity can only be confirmed by an EC measurement.",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-COMP-1",
 "type": "diagnostic",
 "archetype": "ARC-COMPACTED",
 "when": {
  "any": [
   "F-HARD-PACKED",
   "F-SURFACE-POOLING"
  ],
  "also_any": [
   "F-BARE-PATCHES",
   "F-THIN-GRASS"
  ]
 },
 "concludes": "Compaction with depletion from overuse",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-DIST-1",
 "type": "diagnostic",
 "archetype": "ARC-DISTURBED",
 "when": {
  "any": [
   "F-STUMPS"
  ],
  "also_any": [
   "F-NO-CANOPY",
   "F-HARD-PACKED"
  ]
 },
 "concludes": "Disturbed, compacted land with sun exposure as a separate stressor",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-WATER-SAFETY",
 "legacy_name": "R14",
 "type": "safety_filter",
 "when": {
  "water_access": [
   "unknown",
   "none"
  ]
 },
 "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
 "configurable": true,
 "status": "pending_review",
 "condition": {
  "water_access": [
   "unknown",
   "none"
  ]
 },
 "effect_data": {
  "require_water": [
   "low",
   "low-medium"
  ]
 }
},
{
 "id": "R-HEAVY-FEEDER",
 "legacy_name": "R13",
 "type": "safety_filter",
 "when": {
  "deficiency_any": [
   "low organic matter",
   "nutrient depletion",
   "low nutrient retention"
  ]
```

```
  },
  "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
  "configurable": true,
  "status": "pending_review",
  "condition": {
   "archetype_stress_any": [
    "low fertility"
   ]
  },
  "effect_data": {
   "exclude_feeder_class": [
    "heavy"
   ]
  }
 },
 {
  "id": "R-READINESS-GATE",
  "legacy_name": "R9",
  "type": "readiness",
  "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
    counts as not passed.",
  "status": "pending_review"
 },
 {
  "id": "R-MIXED",
  "type": "combination",
  "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
  "status": "pending_review"
 },
 {
  "id": "R-NO-ESTIMATE-AS-FACT",
  "type": "integrity",
  "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
    labelled 'measured'.",
  "status": "pending_review"
 }
],
"procedures": [
 {
  "id": "PROC-ORG-MATTER",
  "name": "Compost / well-rotted manure incorporation",
  "applies_to": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-COMPACTED",
   "ARC-DISTURBED"
  ],
  "targets": [
   "low organic matter",
   "low water retention",
   "nutrient depletion"
  ],
  "prerequisites": [],
  "materials": [
   "compost or well-rotted manure"
  ],
  "rate": 3,
  "unit": "kg per m2",
  "steps": [
   "Spread evenly over the plot.",
   "Incorporate into the topsoil.",
   "Water lightly if water is available."
  ],
  "verification": [
   "Crumb structure test"
  ],
  "stabilization_days": 7,
  "safety": [
   "Use fully rotted material only."
  ],
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
 {
  "id": "PROC-MULCH",
  "name": "Organic mulch cover",
  "applies_to": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-SALINE",
   "ARC-DISTURBED"
  ],
  "targets": [
   "moisture loss",
   "surface protection",
   "wind erosion"
  ],
  "prerequisites": [],
  "materials": [
   "straw, dry grass or crop residue"
  ],
  "rate": 5,
```

```
 "unit": "cm depth",
 "steps": [
  "Apply after amendments are incorporated.",
  "Keep mulch in place against wind (anchor or wet down)."
 ],
 "verification": [
  "Soil under mulch stays visibly moist"
 ],
 "stabilization_days": 0,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-AGRI-GEN"
},
{
 "id": "PROC-BIOCHAR",
 "name": "Biochar incorporation",
 "applies_to": [
  "ARC-SANDY"
 ],
 "targets": [
  "low nutrient retention",
  "low water retention"
 ],
 "prerequisites": [
  "PROC-ORG-MATTER"
 ],
 "materials": [
  "biochar, pre-charged with compost"
 ],
 "rate": 0.5,
 "unit": "kg per m2",
 "steps": [
  "Mix with compost before applying."
 ],
 "verification": [],
 "stabilization_days": 7,
 "phase": "B",
 "status": "pending_review",
 "source": "SRC-AGRI-GEN"
},
{
 "id": "PROC-WINDBREAK",
 "name": "Windbreak and ground cover",
 "applies_to": [
  "ARC-ARID",
  "ARC-SALINE",
  "ARC-SANDY"
 ],
 "targets": [
  "wind erosion",
  "wind damage"
 ],
 "prerequisites": [],
 "materials": [
  "tall hardy plants, brush or woven barriers"
 ],
 "rate": "1 barrier row per 10 m of exposed plot width, across the prevailing wind",
 "unit": "",
 "steps": [
  "Place barriers across the prevailing wind direction.",
  "Cover bare patches before planting."
 ],
 "verification": [
  "Visible dust movement reduced"
 ],
 "stabilization_days": 0,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-CASE"
},
{
 "id": "PROC-DRAINAGE",
 "name": "Drainage correction",
 "applies_to": [
  "ARC-WATERLOGGED"
 ],
 "targets": [
  "standing water",
  "slow drainage"
 ],
 "prerequisites": [],
 "materials": [
  "channels or drains, raised beds"
 ],
 "rate": "Channels about 30 cm wide and 30 cm deep, spaced 5-10 m apart, falling toward an outlet",
 "unit": "",
 "steps": [
  "Dig channels from low patches toward an outlet.",
  "Do not plant until surface water is gone.",
  "Let soil dry until it crumbles rather than smears."
 ],
 "verification": [
  "No standing water",
```

```
  "Infiltration check",
  "Crumb test"
 ],
 "stabilization_days": 7,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-CASE"
},
{
 "id": "PROC-LEACH",
 "name": "Salinity leaching",
 "applies_to": [
  "ARC-SALINE"
 ],
 "targets": [
  "salinity"
 ],
 "prerequisites": [
  "Confirmed low-salt water source",
  "Confirmed drainage out of the root zone"
 ],
 "materials": [
  "low-salt water"
 ],
 "rate": 150,
 "unit": "mm of low-salt water per leaching event; repeat until crust is gone",
 "steps": [
  "Flush with low-salt water only if water and drainage are confirmed."
 ],
 "verification": [
  "EC measurement or crust gone"
 ],
 "stabilization_days": 7,
 "safety": [
  "Leaching without drainage raises the water table and brings salt back up."
 ],
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-AGRI-GEN"
},
{
 "id": "PROC-DECOMPACT",
 "name": "Decompaction and aeration",
 "applies_to": [
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "targets": [
  "compaction",
  "poor infiltration"
 ],
 "prerequisites": [
  "Soil not saturated"
 ],
 "materials": [
  "fork, broadfork or subsoiler"
 ],
 "rate": "Loosen to 25-30 cm depth",
 "unit": "",
 "steps": [
  "Loosen the soil when moderately moist, not wet.",
  "Follow immediately with organic matter."
 ],
 "verification": [
  "Infiltration check"
 ],
 "stabilization_days": 3,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-CASE"
},
{
 "id": "PROC-COVER-CROP",
 "name": "Cover crop / green manure",
 "applies_to": [
  "ARC-COMPACTED",
  "ARC-DISTURBED",
  "ARC-ARID",
  "ARC-SANDY"
 ],
 "targets": [
  "nutrient depletion",
  "bare soil",
  "organic matter"
 ],
 "prerequisites": [
  "PROC-DECOMPACT or PROC-ORG-MATTER where relevant"
 ],
 "materials": [
  "legume cover crop seed"
 ],
 "rate": 5,
 "unit": "g legume seed per m2",
```

```
  "steps": [
   "Sow, grow, then cut and leave or dig in."
  ],
  "verification": [
   "Soil cover established"
  ],
  "stabilization_days": 0,
  "phase": "B",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
 {
  "id": "PROC-SHADE",
  "name": "Temporary shade or surface protection",
  "applies_to": [
   "ARC-DISTURBED",
   "ARC-ARID"
  ],
  "targets": [
   "heat exposure"
  ],
  "prerequisites": [],
  "materials": [
   "shade cloth, mulch or nurse plants"
  ],
  "rate": "30-50% shade cloth or mulch cover over seedlings until established",
  "unit": "",
  "steps": [
   "Protect seedlings during establishment."
  ],
  "verification": [],
  "stabilization_days": 0,
  "phase": "B",
  "status": "pending_review",
  "source": "SRC-CASE"
 }
],
"readiness_criteria": [
 {
  "id": "RC-CRUMB",
  "name": "Crumb structure test",
  "applies_to": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-COMPACTED",
   "ARC-DISTURBED",
   "ARC-WATERLOGGED"
  ],
  "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
    paste.",
  "mandatory": true,
  "if_failed": "Add organic matter or let soil dry, then repeat."
 },
 {
  "id": "RC-INFIL",
  "name": "Infiltration check",
  "applies_to": [
   "ARC-WATERLOGGED",
   "ARC-COMPACTED",
   "ARC-DISTURBED"
  ],
  "check": "Water poured on the surface soaks in without pooling.",
  "mandatory": true,
  "if_failed": "Repeat drainage or decompaction."
 },
 {
  "id": "RC-NOPOOL",
  "name": "No standing water",
  "applies_to": [
   "ARC-WATERLOGGED"
  ],
  "check": "No standing puddles on the plot.",
  "mandatory": true,
  "if_failed": "Continue drainage correction."
 },
 {
  "id": "RC-MOIST",
  "name": "Moisture retention",
  "applies_to": [
   "ARC-SANDY",
   "ARC-ARID"
  ],
  "check": "Soil stays damp below the surface after watering.",
  "mandatory": true,
  "if_failed": "Add organic matter and mulch."
 },
 {
  "id": "RC-DUST",
  "name": "Surface stability",
  "applies_to": [
   "ARC-ARID",
   "ARC-SANDY",
   "ARC-SALINE"
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   ],
   "check": "Wind no longer lifts loose soil from the surface.",
   "mandatory": true,
   "if_failed": "Add windbreak or ground cover."
  },
  {
   "id": "RC-SALT",
   "name": "Salt crust cleared",
   "applies_to": [
    "ARC-SALINE"
   ],
   "check": "No visible crust and a measured EC reading is available.",
   "mandatory": true,
   "if_failed": "Provide an EC reading or continue salinity management."
  },
  {
   "id": "RC-STAB",
   "name": "Stabilization window",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Required waiting time after amendments has passed.",
   "mandatory": true,
   "window_days": 3,
   "if_failed": "Wait for the stabilization window to pass, then repeat."
  }
 ],
 "crops": [
  {
   "id": "CROP-COWPEA",
   "name": "Cowpea / Black-eyed bean",
   "scientific": "Vigna unguiculata",
   "category": "Legume",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility"
   ],
   "avoid": [
    "waterlogging",
    "salinity"
   ],
   "water": "low-medium",
   "light": "full",
   "duration_days": [
    60,
    75
   ],
   "feeder_class": "light",
   "role": "Pioneer crop; fixes nitrogen on poor land.",
   "planting": {
    "depth": "2-4 cm",
    "spacing": "10 cm in row, 40-50 cm between rows",
    "method": "Direct sow into moist, prepared soil"
   },
   "care": {
    "diseases": [
     "Fusarium wilt"
    ],
    "pests": [
     "Aphids",
     "Pod borers"
    ],
    "controls": [
     "Neem-based spray"
    ],
    "fertilizer": "Little nitrogen needed (fixes its own). Compost at bed prep; phosphorus if available."
   },
   "harvest": {
    "indicators": [
     "Pods dry and yellow-brown"
    ],
    "method": "Hand-pick pods as they dry"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "3-7",
    "P2": "20-25",
    "P3": "20-30",
    "P4": "7-10"
   }
  },
  {
   "id": "CROP-SORGHUM",
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 "name": "Sorghum",
 "scientific": "Sorghum bicolor",
 "category": "Cereal / dryland grain",
 "soil_compat": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "heat",
  "marginal soil",
  "some waterlogging",
  "moderate salinity"
 ],
 "avoid": [],
 "water": "low",
 "light": "full",
 "duration_days": [
  90,
  120
 ],
 "feeder_class": "moderate",
 "role": "Drought and marginal-soil tolerant staple grain.",
 "planting": {
  "depth": "2-4 cm",
  "spacing": "15-20 cm in row, 60-75 cm between rows",
  "method": "Direct sow into prepared soil"
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 "care": {
  "diseases": [
   "Grain mold",
   "Anthracnose"
  ],
  "pests": [
   "Shoot fly",
   "Stem borer"
  ],
  "controls": [
   "Neem-based spray",
   "Intercrop with cowpea to disrupt stem borer",
   "Seed treatment with Trichoderma",
   "Reflective ribbon against birds"
  ],
  "fertilizer": "Modest nitrogen and phosphorus; well-rotted compost at bed prep; light top-dress at knee-height
    stage."
 },
 "harvest": {
  "indicators": [
   "Grain hard, heads mature"
  ],
  "method": "Cut heads, dry before storage"
 },
 "status": "pending_review",
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
  "P3": "30-40",
  "P4": "10-14"
 }
},
{
 "id": "CROP-PMILLET",
 "name": "Pearl millet",
 "scientific": "Pennisetum glaucum",
 "category": "Cereal / dryland grain",
 "soil_compat": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-SALINE",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "heat",
  "low fertility",
  "some salinity"
 ],
 "avoid": [
  "waterlogging"
 ],
 "water": "low",
 "light": "full",
 "duration_days": [
  65,
  90
 ],
 "feeder_class": "light",
 "role": "Drought and low-fertility tolerant cereal.",
 "planting": {
  "depth": "2-3 cm",
```

```
  "spacing": "10-15 cm in row, 45-50 cm between rows",
  "method": "Direct sow into prepared soil"
 },
 "care": {
  "diseases": [
   "Downy mildew",
   "Ergot"
  ],
  "pests": [
   "Stem borer"
  ],
  "controls": [
   "Clean or resistant seed",
   "Remove infected plants early",
   "Neem spray"
  ],
  "fertilizer": "Low demand; compost at bed prep and a small nitrogen top-dress at tillering."
 },
 "harvest": {
  "indicators": [
   "Ear heads dry, grains hard"
  ],
  "method": "Cut ear heads and dry"
 },
 "status": "pending_review",
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "3-5",
  "P2": "20-25",
  "P3": "25-35",
  "P4": "7-10"
 }
},
{
 "id": "CROP-AMARANTH",
 "name": "Amaranth",
 "scientific": "Amaranthus spp.",
 "category": "Grain / leafy dual-purpose",
 "soil_compat": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "heat"
 ],
 "avoid": [],
 "water": "low-medium",
 "light": "full",
 "duration_days": [
  30,
  40
 ],
 "duration_note": "Leafy harvest 30-40 days; grain harvest 90-110 days.",
 "feeder_class": "light",
 "role": "Pioneer crop where soil test results are uncertain.",
 "planting": {
  "depth": "0.5-1 cm (tiny seed; mix with sand)",
  "spacing": "10-15 cm in row, 30-45 cm between rows; thin seedlings",
  "method": "Broadcast or drill shallowly into fine, moist soil"
 },
 "care": {
  "diseases": [
   "Damping-off",
   "Leaf spot",
   "Root rot in wet soil"
  ],
  "pests": [
   "Aphids"
  ],
  "controls": [
   "Neem spray",
   "Avoid overwatering",
   "Thin plants for airflow"
  ],
  "fertilizer": "Light feeder; compost at bed prep and a light nitrogen feed after the first leaf cut."
 },
 "harvest": {
  "indicators": [
   "Leaves tender (leafy)",
   "Seed heads dry (grain)"
  ],
  "method": "Cut leaves early or seed heads late"
 },
 "status": "pending_review",
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "3-5",
  "P2": "15-25",
  "P3": "40-60 (grain)",
  "P4": "7-10"
 }
```

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},
{
 "id": "CROP-RICE",
 "name": "Rice (paddy)",
 "scientific": "Oryza sativa",
 "category": "Cereal / wetland",
 "soil_compat": [
  "ARC-WATERLOGGED"
 ],
 "tolerates": [
  "waterlogging"
 ],
 "avoid": [
  "drought"
 ],
 "water": "high",
 "light": "full",
 "duration_days": [
  100,
  150
 ],
 "feeder_class": "moderate",
 "role": "Wet-footed cereal for waterlogged land that cannot be fully drained.",
 "planting": {
  "depth": "2-3 cm; transplant 2-3 seedlings per hill",
  "spacing": "20 x 20 cm",
  "method": "Transplant 20-30 day seedlings into puddled soil with shallow standing water"
 },
 "care": {
  "diseases": [
   "Blast",
   "Sheath blight",
   "Bacterial leaf blight"
  ],
  "pests": [
   "Stem borer",
   "Brown planthopper"
  ],
  "controls": [
   "Resistant varieties",
   "Balanced nitrogen",
   "Brief field drainage to disrupt pests",
   "Neem spray"
  ],
  "fertilizer": "Compost at bed prep; split nitrogen at tillering and panicle initiation."
 },
 "harvest": {
  "indicators": [
   "80-85% of grains straw coloured",
   "Grains hard"
  ],
  "method": "Drain field 7-10 days before; cut, dry, thresh"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
  "P3": "35-45",
  "P4": "10-15"
 }
},
{
 "id": "CROP-TARO",
 "name": "Taro",
 "scientific": "Colocasia esculenta",
 "category": "Root / tuber",
 "soil_compat": [
  "ARC-WATERLOGGED"
 ],
 "tolerates": [
  "waterlogging"
 ],
 "avoid": [
  "drought"
 ],
 "water": "high",
 "light": "full to partial",
 "duration_days": [
  180,
  240
 ],
 "feeder_class": "moderate",
 "role": "Wet-footed root crop.",
 "planting": {
  "depth": "5-10 cm",
  "spacing": "60 x 60 cm",
  "method": "Plant corm pieces or suckers on wet ridges"
 },
 "care": {
  "diseases": [
   "Taro leaf blight",
   "Corm rot"
```

```
   ],
   "pests": [
    "Aphids",
    "Armyworm"
   ],
   "controls": [
    "Remove infected leaves",
    "Use disease-free planting material",
    "Keep wide spacing"
   ],
   "fertilizer": "Compost-heavy; potassium-rich organic matter."
  },
  "harvest": {
   "indicators": [
    "Leaves yellow and die back",
    "Corms firm"
   ],
   "method": "Dig corms carefully; cure before storage"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
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   "P2": "60-90",
   "P3": "60-90",
   "P4": "14-21"
  }
 },
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  "id": "CROP-BARLEY",
  "name": "Barley",
  "scientific": "Hordeum vulgare",
  "category": "Cereal",
  "soil_compat": [
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "tolerates": [
   "salinity",
   "drought"
  ],
  "avoid": [
   "waterlogging"
  ],
  "water": "low-medium",
  "light": "full",
  "duration_days": [
   90,
   120
  ],
  "feeder_class": "moderate",
  "role": "Salt-tolerant cereal.",
  "planting": {
   "depth": "3-5 cm",
   "spacing": "Drill in rows 20 cm apart",
   "method": "Drill or broadcast and cover"
  },
  "care": {
   "diseases": [
    "Powdery mildew",
    "Net blotch",
    "Rust"
   ],
   "pests": [
    "Aphids",
    "Armyworm"
   ],
   "controls": [
    "Resistant varieties",
    "Crop rotation",
    "Neem spray"
   ],
   "fertilizer": "Moderate nitrogen, split; compost at bed prep; avoid chloride-heavy fertilizers."
  },
  "harvest": {
   "indicators": [
    "Heads golden",
    "Grains hard"
   ],
   "method": "Cut when dry, thresh, store below 12% moisture"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
   "P1": "5-10",
   "P2": "30-40",
   "P3": "30-40",
   "P4": "10-14"
  }
 },
 {
  "id": "CROP-PIGEONPEA",
  "name": "Pigeon pea",
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  "scientific": "Cajanus cajan",
  "category": "Legume",
  "soil_compat": [
   "ARC-COMPACTED",
   "ARC-DISTURBED"
  ],
  "tolerates": [
   "drought",
   "compaction (deep roots)"
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  "avoid": [
   "waterlogging"
  ],
  "water": "low",
  "light": "full",
  "duration_days": [
   150,
   180
  ],
  "feeder_class": "light",
  "role": "Deep-rooted legume that helps recover compacted, depleted soil.",
  "planting": {
   "depth": "3-5 cm",
   "spacing": "30 cm in row, 75-90 cm between rows",
   "method": "Direct sow into loosened soil"
  },
  "care": {
   "diseases": [
    "Fusarium wilt",
    "Sterility mosaic",
    "Phytophthora blight"
   ],
   "pests": [
    "Pod borer",
    "Pod fly",
    "Aphids"
   ],
   "controls": [
    "Neem spray",
    "Intercrop with cereals",
    "Resistant varieties",
    "Pheromone traps"
   ],
   "fertilizer": "Little nitrogen; phosphorus helpful; compost at bed prep."
  },
  "harvest": {
   "indicators": [
    "Pods brown and dry",
    "Seeds hard"
   ],
   "method": "Hand-pick or cut dry pods in passes; thresh"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
   "P1": "5-10",
   "P2": "40-60",
   "P3": "50-70",
   "P4": "10-15"
  }
 },
 {
  "id": "CROP-TOMATO",
  "name": "Tomato",
  "scientific": "Solanum lycopersicum",
  "category": "Vegetable",
  "soil_compat": [],
  "tolerates": [],
  "avoid": [
   "drought",
   "low fertility",
   "salinity"
  ],
  "water": "high",
  "light": "full",
  "duration_days": [
   90,
   120
  ],
  "feeder_class": "heavy",
  "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
  "planting": {
   "depth": "0.5-1 cm in nursery; transplant at 5 cm",
   "spacing": "45 x 60 cm",
   "method": "Raise seedlings, then transplant"
  },
  "care": {
   "diseases": [
    "Early blight",
    "Late blight",
    "Bacterial wilt"
   ],
   "pests": [
```

```
    "Whitefly",
    "Fruit borer"
   ],
   "controls": [
    "Neem spray",
    "Crop rotation",
    "Stake and prune for airflow"
   ],
   "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
  },
  "harvest": {
   "indicators": [
    "Fruit fully coloured and slightly firm"
   ],
   "method": "Hand-pick ripe fruit regularly"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
   "P1": "5-8",
   "P2": "25-30",
   "P3": "30-45",
   "P4": "7-14"
  }
 }
],
"phases": {
 "journey": [
  {
   "id": "P1",
   "name": "Germination"
  },
  {
   "id": "P2",
   "name": "Vegetative Growth"
  },
  {
   "id": "P3",
   "name": "Reproductive & Canopy Development"
  },
  {
   "id": "P4",
   "name": "Pre-Harvest"
  }
 ],
 "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
    data is unavailable."
},
"missing_data_messages": {
 "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
 "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
 "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
 "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
 "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
},
"scenarios": [
 {
  "id": "SC-1",
  "name": "Arid Wasteland",
  "expect_archetypes": [
   "ARC-ARID",
   "ARC-SANDY"
  ],
  "expect_features": [
   "F-CRACKED",
   "F-LOOSE-DUST",
   "F-NO-VEG",
   "F-HIGH-HEAT",
   "F-RAIN-RARE"
  ]
 },
 {
  "id": "SC-2",
  "name": "Flood-Damaged Wetland",
  "expect_archetypes": [
   "ARC-WATERLOGGED"
  ],
  "expect_features": [
   "F-DARK-STICKY",
   "F-STANDING-WATER",
   "F-SOUR-SMELL",
   "F-YELLOW-WEEDS",
   "F-SLOW-DRAIN"
  ]
 },
 {
  "id": "SC-3",
  "name": "Coastal Saline Land",
  "expect_archetypes": [
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "expect_features": [
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    "F-GRITTY",
    "F-SALT-CRUST",
    "F-SALT-SMELL",
    "F-STUNTED-BROWN",
    "F-SEA-NEAR"
   ]
  },
  {
   "id": "SC-4",
   "name": "Overgrazed Pastureland",
   "expect_archetypes": [
    "ARC-COMPACTED"
   ],
   "expect_features": [
    "F-HARD-PACKED",
    "F-SURFACE-POOLING",
    "F-BARE-PATCHES",
    "F-THIN-GRASS"
   ]
  },
  {
   "id": "SC-5",
   "name": "Post-Deforestation Cleared Land",
   "expect_archetypes": [
    "ARC-DISTURBED"
   ],
   "expect_features": [
    "F-STUMPS",
    "F-NO-CANOPY",
    "F-HARD-PACKED"
   ]
  }
 ]
}
```

3. kb/case1_arid.json

Full file contents


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{
 "meta": {
  "name": "TERRAFORM KB - case1_arid",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ],
  "water_confirm_features": [
   "F-STANDING-WATER"
  ],
  "conflicts": [
   [
    "F-STANDING-WATER",
    "F-CRACKED"
   ],
   [
    "F-STANDING-WATER",
    "F-RAIN-RARE"
   ]
  ],
  "text_limit": 3000
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 "sources": [
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   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
  },
  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
  },
  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
 ],
 "features": [
  {
   "id": "F-CRACKED",
   "label": "Cracked soil surface",
   "kind": "visual",
   "lexicon": "crack|plates|fissur"
  },
  {
   "id": "F-LOOSE-DUST",
   "label": "Loose soil or dust lifted by wind",
   "kind": "visual",
   "lexicon": "dust|wind[- ]?(blown|lifting)|dust|lifting loose|loose soil|erosion|eroding"
  },
  {
   "id": "F-GRITTY",
   "label": "Gritty, loose, sandy texture",
   "kind": "text",
   "lexicon": "grit|sandy|\\bsand\\b|beach"
  },
  {
   "id": "F-NO-VEG",
   "label": "No or very sparse vegetation",
   "kind": "visual",
   "lexicon": "no (visible )?(veg|plants|green)|barren|bare (land|plot|ground)|sparse|scrap of green|dead
     (veg|weeds)|brittle weeds"
  },
  {
   "id": "F-HIGH-HEAT",
   "label": "Daytime temperatures reported above 40 C",
   "kind": "text",
   "lexicon": "\\b4\\d ?\u00b0? ?c|scorching|extreme heat|very hot|intense heat"
  },
  {
   "id": "F-RAIN-RARE",
   "label": "Rainfall rare or unreliable",
   "kind": "text",
   "lexicon": "rain(fall)? (is )?(rare|unreliable|unpredictable|scarce)|rare.{0,30}rain|no rain|last rain"
  }
 ],
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"toggles": [
 {
  "id": "T-WATER",
  "label": "Water Source Nearby",
  "sets": {
   "water_access": "confirmed"
  }
 },
 {
  "id": "T-SLOPE",
  "label": "Sloped Terrain",
  "sets": {
   "slope": true
  }
 },
 {
  "id": "T-ROCKY",
  "label": "Rocky",
  "sets": {
   "rocky": true
  }
 },
 {
  "id": "T-SHADE",
  "label": "Shaded",
  "sets": {
   "shaded": true
  },
  "cancels_features": [
   "F-NO-CANOPY"
  ]
 },
 {
  "id": "T-RAIN",
  "label": "Recent Rain",
  "sets": {
   "recent_rain": true
  }
 },
 {
  "id": "T-WIND",
  "label": "Strong Wind",
  "sets": {
   "strong_wind": true
  }
 },
 {
  "id": "T-SALT-AIR",
  "label": "Salty/Coastal Air",
  "sets": {
   "salt_air": true
  },
  "adds_features": [
   "F-SALT-SMELL"
  ]
 }
],
"archetypes": [
 {
  "id": "ARC-SANDY",
  "name": "Sandy / Loamy Sand (Degraded)",
  "criteria": "Gritty, loose texture; low cohesion; rapid drainage.",
  "physical_profile": "Loose, particulate, low aggregate stability",
  "core_gaps": [
   "low water-holding capacity",
   "low organic matter",
   "low nutrient retention"
  ],
  "typical_estimates": {
   "texture_class": "Sand / loamy sand",
   "drainage": "Excessive",
   "ph_range": "5.5-7.0",
   "organic_matter": "Low",
   "cec": "Low"
  },
  "estimate_label": "estimate, not laboratory measurement",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "stress_tags": [
   "drought",
   "low fertility"
  ]
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 {
  "id": "ARC-ARID",
  "name": "Arid / Drought-Stressed",
  "criteria": "Dry, low organic matter, poor moisture retention, unreliable rain.",
  "variants": [
   "cracked-compacted (retention dominant)",
   "loose-eroding (wind erosion dominant)"
  ],
  "core_gaps": [
   "water scarcity",
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    "wind erosion",
    "poor structure"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "stress_tags": [
    "drought",
    "heat",
    "low fertility"
   ],
   "typical_estimates": {
    "texture_class": "Clay-loam crust to loose sand",
    "drainage": "Poor when crusted, rapid when sandy",
    "organic_matter": "Low",
    "cec": "Low to moderate",
    "ph_range": "7.0-8.5"
   }
  }
 ],
 "rules": [
  {
   "id": "R-ARID-1",
   "type": "diagnostic",
   "archetype": "ARC-ARID",
   "when": {
    "all": [
     "F-NO-VEG"
    ],
    "any": [
     "F-CRACKED",
     "F-RAIN-RARE",
     "F-HIGH-HEAT"
    ]
   },
   "concludes": "Drought stress with poor moisture retention",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-ARID-2",
   "type": "diagnostic",
   "archetype": "ARC-ARID",
   "when": {
    "all": [
     "F-LOOSE-DUST"
    ],
    "any": [
     "F-NO-VEG",
     "F-GRITTY",
     "F-CRACKED"
    ]
   },
   "concludes": "Active wind erosion in addition to dryness",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-SANDY-1",
   "type": "diagnostic",
   "archetype": "ARC-SANDY",
   "when": {
    "all": [
     "F-GRITTY"
    ]
   },
   "concludes": "Sandy texture. Texture is not by itself degradation; confirm low organic matter before diagnosing
     degradation.",
   "min_evidence": 1,
   "status": "pending_review"
  },
  {
   "id": "R-WATER-SAFETY",
   "legacy_name": "R14",
   "type": "safety_filter",
   "when": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect_data": {
    "require_water": [
     "low",
     "low-medium"
```

```
    ]
   }
  },
  {
   "id": "R-HEAVY-FEEDER",
   "legacy_name": "R13",
   "type": "safety_filter",
   "when": {
    "deficiency_any": [
     "low organic matter",
     "nutrient depletion",
     "low nutrient retention"
    ]
   },
   "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "archetype_stress_any": [
     "low fertility"
    ]
   },
   "effect_data": {
    "exclude_feeder_class": [
     "heavy"
    ]
   }
  },
  {
   "id": "R-READINESS-GATE",
   "legacy_name": "R9",
   "type": "readiness",
   "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
     counts as not passed.",
   "status": "pending_review"
  },
  {
   "id": "R-MIXED",
   "type": "combination",
   "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
   "status": "pending_review"
  },
  {
   "id": "R-NO-ESTIMATE-AS-FACT",
   "type": "integrity",
   "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
     labelled 'measured'.",
   "status": "pending_review"
  }
 ],
 "procedures": [
  {
   "id": "PROC-ORG-MATTER",
   "name": "Compost / well-rotted manure incorporation",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "targets": [
    "low organic matter",
    "low water retention",
    "nutrient depletion"
   ],
   "prerequisites": [],
   "materials": [
    "compost or well-rotted manure"
   ],
   "rate": 3,
   "unit": "kg per m2",
   "steps": [
    "Spread evenly over the plot.",
    "Incorporate into the topsoil.",
    "Water lightly if water is available."
   ],
   "verification": [
    "Crumb structure test"
   ],
   "stabilization_days": 7,
   "safety": [
    "Use fully rotted material only."
   ],
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-MULCH",
   "name": "Organic mulch cover",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
```

```
   "ARC-SALINE",
   "ARC-DISTURBED"
  ],
  "targets": [
   "moisture loss",
   "surface protection",
   "wind erosion"
  ],
  "prerequisites": [],
  "materials": [
   "straw, dry grass or crop residue"
  ],
  "rate": 5,
  "unit": "cm depth",
  "steps": [
   "Apply after amendments are incorporated.",
   "Keep mulch in place against wind (anchor or wet down)."
  ],
  "verification": [
   "Soil under mulch stays visibly moist"
  ],
  "stabilization_days": 0,
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
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 {
  "id": "PROC-BIOCHAR",
  "name": "Biochar incorporation",
  "applies_to": [
   "ARC-SANDY"
  ],
  "targets": [
   "low nutrient retention",
   "low water retention"
  ],
  "prerequisites": [
   "PROC-ORG-MATTER"
  ],
  "materials": [
   "biochar, pre-charged with compost"
  ],
  "rate": 0.5,
  "unit": "kg per m2",
  "steps": [
   "Mix with compost before applying."
  ],
  "verification": [],
  "stabilization_days": 7,
  "phase": "B",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
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  "id": "PROC-WINDBREAK",
  "name": "Windbreak and ground cover",
  "applies_to": [
   "ARC-ARID",
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "targets": [
   "wind erosion",
   "wind damage"
  ],
  "prerequisites": [],
  "materials": [
   "tall hardy plants, brush or woven barriers"
  ],
  "rate": "1 barrier row per 10 m of exposed plot width, across the prevailing wind",
  "unit": "",
  "steps": [
   "Place barriers across the prevailing wind direction.",
   "Cover bare patches before planting."
  ],
  "verification": [
   "Visible dust movement reduced"
  ],
  "stabilization_days": 0,
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-CASE"
 },
 {
  "id": "PROC-COVER-CROP",
  "name": "Cover crop / green manure",
  "applies_to": [
   "ARC-COMPACTED",
   "ARC-DISTURBED",
   "ARC-ARID",
   "ARC-SANDY"
  ],
  "targets": [
```

```
    "nutrient depletion",
    "bare soil",
    "organic matter"
   ],
   "prerequisites": [
    "PROC-DECOMPACT or PROC-ORG-MATTER where relevant"
   ],
   "materials": [
    "legume cover crop seed"
   ],
   "rate": 5,
   "unit": "g legume seed per m2",
   "steps": [
    "Sow, grow, then cut and leave or dig in."
   ],
   "verification": [
    "Soil cover established"
   ],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-SHADE",
   "name": "Temporary shade or surface protection",
   "applies_to": [
    "ARC-DISTURBED",
    "ARC-ARID"
   ],
   "targets": [
    "heat exposure"
   ],
   "prerequisites": [],
   "materials": [
    "shade cloth, mulch or nurse plants"
   ],
   "rate": "30-50% shade cloth or mulch cover over seedlings until established",
   "unit": "",
   "steps": [
    "Protect seedlings during establishment."
   ],
   "verification": [],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-CASE"
  }
 ],
 "readiness_criteria": [
  {
   "id": "RC-CRUMB",
   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-MOIST",
   "name": "Moisture retention",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID"
   ],
   "check": "Soil stays damp below the surface after watering.",
   "mandatory": true,
   "if_failed": "Add organic matter and mulch."
  },
  {
   "id": "RC-DUST",
   "name": "Surface stability",
   "applies_to": [
    "ARC-ARID",
    "ARC-SANDY",
    "ARC-SALINE"
   ],
   "check": "Wind no longer lifts loose soil from the surface.",
   "mandatory": true,
   "if_failed": "Add windbreak or ground cover."
  },
  {
   "id": "RC-STAB",
   "name": "Stabilization window",
   "applies_to": [
    "ARC-SANDY",
```

```
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Required waiting time after amendments has passed.",
   "mandatory": true,
   "window_days": 3,
   "if_failed": "Wait for the stabilization window to pass, then repeat."
  }
 ],
 "crops": [
  {
   "id": "CROP-COWPEA",
   "name": "Cowpea / Black-eyed bean",
   "scientific": "Vigna unguiculata",
   "category": "Legume",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility"
   ],
   "avoid": [
    "waterlogging",
    "salinity"
   ],
   "water": "low-medium",
   "light": "full",
   "duration_days": [
    60,
    75
   ],
   "feeder_class": "light",
   "role": "Pioneer crop; fixes nitrogen on poor land.",
   "planting": {
    "depth": "2-4 cm",
    "spacing": "10 cm in row, 40-50 cm between rows",
    "method": "Direct sow into moist, prepared soil"
   },
   "care": {
    "diseases": [
     "Fusarium wilt"
    ],
    "pests": [
     "Aphids",
     "Pod borers"
    ],
    "controls": [
     "Neem-based spray"
    ],
    "fertilizer": "Little nitrogen needed (fixes its own). Compost at bed prep; phosphorus if available."
   },
   "harvest": {
    "indicators": [
     "Pods dry and yellow-brown"
    ],
    "method": "Hand-pick pods as they dry"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "3-7",
    "P2": "20-25",
    "P3": "20-30",
    "P4": "7-10"
   }
  },
  {
   "id": "CROP-SORGHUM",
   "name": "Sorghum",
   "scientific": "Sorghum bicolor",
   "category": "Cereal / dryland grain",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "marginal soil",
    "some waterlogging",
    "moderate salinity"
   ],
   "avoid": [],
   "water": "low",
   "light": "full",
   "duration_days": [
```

```
    90,
    120
   ],
   "feeder_class": "moderate",
   "role": "Drought and marginal-soil tolerant staple grain.",
   "planting": {
    "depth": "2-4 cm",
    "spacing": "15-20 cm in row, 60-75 cm between rows",
    "method": "Direct sow into prepared soil"
   },
   "care": {
    "diseases": [
     "Grain mold",
     "Anthracnose"
    ],
    "pests": [
     "Shoot fly",
     "Stem borer"
    ],
    "controls": [
     "Neem-based spray",
     "Intercrop with cowpea to disrupt stem borer",
     "Seed treatment with Trichoderma",
     "Reflective ribbon against birds"
    ],
    "fertilizer": "Modest nitrogen and phosphorus; well-rotted compost at bed prep; light top-dress at knee-height
      stage."
   },
   "harvest": {
    "indicators": [
     "Grain hard, heads mature"
    ],
    "method": "Cut heads, dry before storage"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "5-10",
    "P2": "30-40",
    "P3": "30-40",
    "P4": "10-14"
   }
  },
  {
   "id": "CROP-PMILLET",
   "name": "Pearl millet",
   "scientific": "Pennisetum glaucum",
   "category": "Cereal / dryland grain",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-SALINE",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility",
    "some salinity"
   ],
   "avoid": [
    "waterlogging"
   ],
   "water": "low",
   "light": "full",
   "duration_days": [
    65,
    90
   ],
   "feeder_class": "light",
   "role": "Drought and low-fertility tolerant cereal.",
   "planting": {
    "depth": "2-3 cm",
    "spacing": "10-15 cm in row, 45-50 cm between rows",
    "method": "Direct sow into prepared soil"
   },
   "care": {
    "diseases": [
     "Downy mildew",
     "Ergot"
    ],
    "pests": [
     "Stem borer"
    ],
    "controls": [
     "Clean or resistant seed",
     "Remove infected plants early",
     "Neem spray"
    ],
    "fertilizer": "Low demand; compost at bed prep and a small nitrogen top-dress at tillering."
   },
   "harvest": {
    "indicators": [
```

```
    "Ear heads dry, grains hard"
   ],
   "method": "Cut ear heads and dry"
  },
  "status": "pending_review",
  "source": "SRC-PRD",
  "phase_days": {
   "P1": "3-5",
   "P2": "20-25",
   "P3": "25-35",
   "P4": "7-10"
  }
 },
 {
  "id": "CROP-AMARANTH",
  "name": "Amaranth",
  "scientific": "Amaranthus spp.",
  "category": "Grain / leafy dual-purpose",
  "soil_compat": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-DISTURBED"
  ],
  "tolerates": [
   "drought",
   "heat"
  ],
  "avoid": [],
  "water": "low-medium",
  "light": "full",
  "duration_days": [
   30,
   40
  ],
  "duration_note": "Leafy harvest 30-40 days; grain harvest 90-110 days.",
  "feeder_class": "light",
  "role": "Pioneer crop where soil test results are uncertain.",
  "planting": {
   "depth": "0.5-1 cm (tiny seed; mix with sand)",
   "spacing": "10-15 cm in row, 30-45 cm between rows; thin seedlings",
   "method": "Broadcast or drill shallowly into fine, moist soil"
  },
  "care": {
   "diseases": [
    "Damping-off",
    "Leaf spot",
    "Root rot in wet soil"
   ],
   "pests": [
    "Aphids"
   ],
   "controls": [
    "Neem spray",
    "Avoid overwatering",
    "Thin plants for airflow"
   ],
   "fertilizer": "Light feeder; compost at bed prep and a light nitrogen feed after the first leaf cut."
  },
  "harvest": {
   "indicators": [
    "Leaves tender (leafy)",
    "Seed heads dry (grain)"
   ],
   "method": "Cut leaves early or seed heads late"
  },
  "status": "pending_review",
  "source": "SRC-PRD",
  "phase_days": {
   "P1": "3-5",
   "P2": "15-25",
   "P3": "40-60 (grain)",
   "P4": "7-10"
  }
 },
 {
  "id": "CROP-TOMATO",
  "name": "Tomato",
  "scientific": "Solanum lycopersicum",
  "category": "Vegetable",
  "soil_compat": [],
  "tolerates": [],
  "avoid": [
   "drought",
   "low fertility",
   "salinity"
  ],
  "water": "high",
  "light": "full",
  "duration_days": [
   90,
   120
  ],
  "feeder_class": "heavy",
```

```
   "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
   "planting": {
    "depth": "0.5-1 cm in nursery; transplant at 5 cm",
    "spacing": "45 x 60 cm",
    "method": "Raise seedlings, then transplant"
   },
   "care": {
    "diseases": [
     "Early blight",
     "Late blight",
     "Bacterial wilt"
    ],
    "pests": [
     "Whitefly",
     "Fruit borer"
    ],
    "controls": [
     "Neem spray",
     "Crop rotation",
     "Stake and prune for airflow"
    ],
    "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
   },
   "harvest": {
    "indicators": [
     "Fruit fully coloured and slightly firm"
    ],
    "method": "Hand-pick ripe fruit regularly"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
    "P1": "5-8",
    "P2": "25-30",
    "P3": "30-45",
    "P4": "7-14"
   }
  }
 ],
 "phases": {
  "journey": [
   {
    "id": "P1",
    "name": "Germination"
   },
   {
    "id": "P2",
    "name": "Vegetative Growth"
   },
   {
    "id": "P3",
    "name": "Reproductive & Canopy Development"
   },
   {
    "id": "P4",
    "name": "Pre-Harvest"
   }
  ],
  "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
     data is unavailable."
 },
 "missing_data_messages": {
  "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
  "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
  "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
  "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
  "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
 }
}
```

4. kb/case2_wetland.json

Full file contents


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{
 "meta": {
  "name": "TERRAFORM KB - case2_wetland",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ],
  "water_confirm_features": [
   "F-STANDING-WATER"
  ],
  "conflicts": [
   [
    "F-STANDING-WATER",
    "F-CRACKED"
   ],
   [
    "F-STANDING-WATER",
    "F-RAIN-RARE"
   ]
  ],
  "text_limit": 3000
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 "sources": [
  {
   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
  },
  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
  },
  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
 ],
 "features": [
  {
   "id": "F-STANDING-WATER",
   "label": "Standing water or puddles",
   "kind": "visual",
   "lexicon": "standing (water|puddle)|puddle|pooling|submerged|flooded"
  },
  {
   "id": "F-SOUR-SMELL",
   "label": "Sour or musty soil smell",
   "kind": "text",
   "lexicon": "sour|musty|foul smell"
  },
  {
   "id": "F-YELLOW-WEEDS",
   "label": "Yellowing weeds with soft or rotting roots",
   "kind": "visual",
   "lexicon": "yellow|rotting roots|soft roots|wilting"
  },
  {
   "id": "F-SLOW-DRAIN",
   "label": "Drains slowly, stays wet after rain",
   "kind": "text",
   "lexicon": "slow(ly)? drain|drains slowly|stays wet|waterlog|poorly drain"
  },
  {
   "id": "F-DARK-STICKY",
   "label": "Dark, sticky clay",
   "kind": "text",
   "lexicon": "sticky|dark clay|heavy clay"
  }
 ],
 "toggles": [
  {
   "id": "T-WATER",
   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
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 },
 {
  "id": "T-SLOPE",
  "label": "Sloped Terrain",
  "sets": {
   "slope": true
  }
 },
 {
  "id": "T-ROCKY",
  "label": "Rocky",
  "sets": {
   "rocky": true
  }
 },
 {
  "id": "T-SHADE",
  "label": "Shaded",
  "sets": {
   "shaded": true
  },
  "cancels_features": [
   "F-NO-CANOPY"
  ]
 },
 {
  "id": "T-RAIN",
  "label": "Recent Rain",
  "sets": {
   "recent_rain": true
  }
 },
 {
  "id": "T-WIND",
  "label": "Strong Wind",
  "sets": {
   "strong_wind": true
  }
 },
 {
  "id": "T-SALT-AIR",
  "label": "Salty/Coastal Air",
  "sets": {
   "salt_air": true
  },
  "adds_features": [
   "F-SALT-SMELL"
  ]
 }
],
"archetypes": [
 {
  "id": "ARC-WATERLOGGED",
  "name": "Waterlogged / Poorly Drained",
  "criteria": "Standing water or slow drainage; sour smell; oxygen-starved root zone.",
  "variants": [
   "recently flooded (days)",
   "prolonged waterlogging",
   "heavy slow-draining without visible pooling"
  ],
  "core_gaps": [
   "drainage",
   "root-zone oxygen",
   "nutrient loss",
   "disease risk"
  ],
  "status": "pending_review",
  "source": "SRC-CASE",
  "stress_tags": [
   "waterlogging"
  ],
  "typical_estimates": {
   "texture_class": "Clay / silty clay",
   "drainage": "Very poor",
   "organic_matter": "Moderate, partly decomposed",
   "cec": "Moderate to high",
   "ph_range": "5.5-6.5"
  }
 }
],
"rules": [
 {
  "id": "R-WET-1",
  "type": "diagnostic",
  "archetype": "ARC-WATERLOGGED",
  "when": {
   "any": [
    "F-STANDING-WATER",
    "F-SLOW-DRAIN"
   ],
   "also_any": [
    "F-SOUR-SMELL",
    "F-YELLOW-WEEDS",
```

```
      "F-DARK-STICKY"
     ]
    },
    "concludes": "Waterlogging with root-zone oxygen stress",
    "min_evidence": 2,
    "status": "pending_review"
   },
   {
    "id": "R-WATER-SAFETY",
    "legacy_name": "R14",
    "type": "safety_filter",
    "when": {
     "water_access": [
      "unknown",
      "none"
     ]
    },
    "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
    "configurable": true,
    "status": "pending_review",
    "condition": {
     "water_access": [
      "unknown",
      "none"
     ]
    },
    "effect_data": {
     "require_water": [
      "low",
      "low-medium"
     ]
    }
   },
   {
    "id": "R-HEAVY-FEEDER",
    "legacy_name": "R13",
    "type": "safety_filter",
    "when": {
     "deficiency_any": [
      "low organic matter",
      "nutrient depletion",
      "low nutrient retention"
     ]
    },
    "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
    "configurable": true,
    "status": "pending_review",
    "condition": {
     "archetype_stress_any": [
      "low fertility"
     ]
    },
    "effect_data": {
     "exclude_feeder_class": [
      "heavy"
     ]
    }
   },
   {
    "id": "R-READINESS-GATE",
    "legacy_name": "R9",
    "type": "readiness",
    "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
      counts as not passed.",
    "status": "pending_review"
   },
   {
    "id": "R-MIXED",
    "type": "combination",
    "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
    "status": "pending_review"
   },
   {
    "id": "R-NO-ESTIMATE-AS-FACT",
    "type": "integrity",
    "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
      labelled 'measured'.",
    "status": "pending_review"
   }
  ],
  "procedures": [
   {
    "id": "PROC-DRAINAGE",
    "name": "Drainage correction",
    "applies_to": [
     "ARC-WATERLOGGED"
    ],
    "targets": [
     "standing water",
     "slow drainage"
    ],
    "prerequisites": [],
    "materials": [
```

```
    "channels or drains, raised beds"
   ],
   "rate": "Channels about 30 cm wide and 30 cm deep, spaced 5-10 m apart, falling toward an outlet",
   "unit": "",
   "steps": [
    "Dig channels from low patches toward an outlet.",
    "Do not plant until surface water is gone.",
    "Let soil dry until it crumbles rather than smears."
   ],
   "verification": [
    "No standing water",
    "Infiltration check",
    "Crumb test"
   ],
   "stabilization_days": 7,
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-CASE"
  }
 ],
 "readiness_criteria": [
  {
   "id": "RC-CRUMB",
   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-INFIL",
   "name": "Infiltration check",
   "applies_to": [
    "ARC-WATERLOGGED",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Water poured on the surface soaks in without pooling.",
   "mandatory": true,
   "if_failed": "Repeat drainage or decompaction."
  },
  {
   "id": "RC-NOPOOL",
   "name": "No standing water",
   "applies_to": [
    "ARC-WATERLOGGED"
   ],
   "check": "No standing puddles on the plot.",
   "mandatory": true,
   "if_failed": "Continue drainage correction."
  }
 ],
 "crops": [
  {
   "id": "CROP-RICE",
   "name": "Rice (paddy)",
   "scientific": "Oryza sativa",
   "category": "Cereal / wetland",
   "soil_compat": [
    "ARC-WATERLOGGED"
   ],
   "tolerates": [
    "waterlogging"
   ],
   "avoid": [
    "drought"
   ],
   "water": "high",
   "light": "full",
   "duration_days": [
    100,
    150
   ],
   "feeder_class": "moderate",
   "role": "Wet-footed cereal for waterlogged land that cannot be fully drained.",
   "planting": {
    "depth": "2-3 cm; transplant 2-3 seedlings per hill",
    "spacing": "20 x 20 cm",
    "method": "Transplant 20-30 day seedlings into puddled soil with shallow standing water"
   },
   "care": {
    "diseases": [
     "Blast",
     "Sheath blight",
     "Bacterial leaf blight"
    ],
```

```
  "pests": [
   "Stem borer",
   "Brown planthopper"
  ],
  "controls": [
   "Resistant varieties",
   "Balanced nitrogen",
   "Brief field drainage to disrupt pests",
   "Neem spray"
  ],
  "fertilizer": "Compost at bed prep; split nitrogen at tillering and panicle initiation."
 },
 "harvest": {
  "indicators": [
   "80-85% of grains straw coloured",
   "Grains hard"
  ],
  "method": "Drain field 7-10 days before; cut, dry, thresh"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
  "P3": "35-45",
  "P4": "10-15"
 }
},
{
 "id": "CROP-TARO",
 "name": "Taro",
 "scientific": "Colocasia esculenta",
 "category": "Root / tuber",
 "soil_compat": [
  "ARC-WATERLOGGED"
 ],
 "tolerates": [
  "waterlogging"
 ],
 "avoid": [
  "drought"
 ],
 "water": "high",
 "light": "full to partial",
 "duration_days": [
  180,
  240
 ],
 "feeder_class": "moderate",
 "role": "Wet-footed root crop.",
 "planting": {
  "depth": "5-10 cm",
  "spacing": "60 x 60 cm",
  "method": "Plant corm pieces or suckers on wet ridges"
 },
 "care": {
  "diseases": [
   "Taro leaf blight",
   "Corm rot"
  ],
  "pests": [
   "Aphids",
   "Armyworm"
  ],
  "controls": [
   "Remove infected leaves",
   "Use disease-free planting material",
   "Keep wide spacing"
  ],
  "fertilizer": "Compost-heavy; potassium-rich organic matter."
 },
 "harvest": {
  "indicators": [
   "Leaves yellow and die back",
   "Corms firm"
  ],
  "method": "Dig corms carefully; cure before storage"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "14-21",
  "P2": "60-90",
  "P3": "60-90",
  "P4": "14-21"
 }
},
{
 "id": "CROP-TOMATO",
 "name": "Tomato",
 "scientific": "Solanum lycopersicum",
 "category": "Vegetable",
 "soil_compat": [],
```

```
    "tolerates": [],
    "avoid": [
     "drought",
     "low fertility",
     "salinity"
    ],
    "water": "high",
    "light": "full",
    "duration_days": [
     90,
     120
    ],
    "feeder_class": "heavy",
    "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
    "planting": {
     "depth": "0.5-1 cm in nursery; transplant at 5 cm",
     "spacing": "45 x 60 cm",
     "method": "Raise seedlings, then transplant"
    },
    "care": {
     "diseases": [
      "Early blight",
      "Late blight",
      "Bacterial wilt"
     ],
     "pests": [
      "Whitefly",
      "Fruit borer"
     ],
     "controls": [
      "Neem spray",
      "Crop rotation",
      "Stake and prune for airflow"
     ],
     "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
    },
    "harvest": {
     "indicators": [
      "Fruit fully coloured and slightly firm"
     ],
     "method": "Hand-pick ripe fruit regularly"
    },
    "status": "pending_review",
    "source": "SRC-AGRI-GEN",
    "phase_days": {
     "P1": "5-8",
     "P2": "25-30",
     "P3": "30-45",
     "P4": "7-14"
    }
   }
  ],
  "phases": {
   "journey": [
    {
     "id": "P1",
     "name": "Germination"
    },
    {
     "id": "P2",
     "name": "Vegetative Growth"
    },
    {
     "id": "P3",
     "name": "Reproductive & Canopy Development"
    },
    {
     "id": "P4",
     "name": "Pre-Harvest"
    }
   ],
   "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
      data is unavailable."
  },
  "missing_data_messages": {
   "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
   "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
   "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
   "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
   "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
  }
}
```

5. kb/case3_saline.json

Full file contents


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{
 "meta": {
  "name": "TERRAFORM KB - case3_saline",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ],
  "water_confirm_features": [
   "F-STANDING-WATER"
  ],
  "conflicts": [
   [
    "F-STANDING-WATER",
    "F-CRACKED"
   ],
   [
    "F-STANDING-WATER",
    "F-RAIN-RARE"
   ]
  ],
  "text_limit": 3000
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 "sources": [
  {
   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
  },
  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
  },
  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
 ],
 "features": [
  {
   "id": "F-GRITTY",
   "label": "Gritty, loose, sandy texture",
   "kind": "text",
   "lexicon": "grit|sandy|\\bsand\\b|beach"
  },
  {
   "id": "F-SALT-CRUST",
   "label": "White or pale crust on surface",
   "kind": "visual",
   "lexicon": "white.{0,25}(crust|residue)|salt crust|crusty|crusted"
  },
  {
   "id": "F-SALT-SMELL",
   "label": "Salty air or smell",
   "kind": "text",
   "lexicon": "salt(y)? (smell|air|wind|spray)|smell of salt|salty"
  },
  {
   "id": "F-STUNTED-BROWN",
   "label": "Stunted, browning vegetation",
   "kind": "visual",
   "lexicon": "stunted|browning|scorched"
  },
  {
   "id": "F-SEA-NEAR",
   "label": "Sea or shoreline nearby",
   "kind": "text",
   "lexicon": "\\bsea\\b|shore|coast|ocean"
  }
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 "toggles": [
  {
   "id": "T-WATER",
   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
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```
  },
  {
   "id": "T-SLOPE",
   "label": "Sloped Terrain",
   "sets": {
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   }
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   "id": "T-ROCKY",
   "label": "Rocky",
   "sets": {
    "rocky": true
   }
  },
  {
   "id": "T-SHADE",
   "label": "Shaded",
   "sets": {
    "shaded": true
   },
   "cancels_features": [
    "F-NO-CANOPY"
   ]
  },
  {
   "id": "T-RAIN",
   "label": "Recent Rain",
   "sets": {
    "recent_rain": true
   }
  },
  {
   "id": "T-WIND",
   "label": "Strong Wind",
   "sets": {
    "strong_wind": true
   }
  },
  {
   "id": "T-SALT-AIR",
   "label": "Salty/Coastal Air",
   "sets": {
    "salt_air": true
   },
   "adds_features": [
    "F-SALT-SMELL"
   ]
  }
 ],
 "archetypes": [
  {
   "id": "ARC-SANDY",
   "name": "Sandy / Loamy Sand (Degraded)",
   "criteria": "Gritty, loose texture; low cohesion; rapid drainage.",
   "physical_profile": "Loose, particulate, low aggregate stability",
   "core_gaps": [
    "low water-holding capacity",
    "low organic matter",
    "low nutrient retention"
   ],
   "typical_estimates": {
    "texture_class": "Sand / loamy sand",
    "drainage": "Excessive",
    "ph_range": "5.5-7.0",
    "organic_matter": "Low",
    "cec": "Low"
   },
   "estimate_label": "estimate, not laboratory measurement",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "stress_tags": [
    "drought",
    "low fertility"
   ]
  },
  {
   "id": "ARC-SALINE",
   "name": "Saline / Salt-Affected",
   "criteria": "Salt crust or salty air, stunted browning plants, coastal context.",
   "variants": [
    "visible crust",
    "no crust but salty smell and stunting",
    "salt spray without saline soil"
   ],
   "core_gaps": [
    "salinity",
    "wind exposure",
    "low water retention if sandy"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "stress_tags": [
```

```
    "salinity"
   ],
   "typical_estimates": {
    "texture_class": "Sand / loamy sand",
    "drainage": "Rapid",
    "organic_matter": "Low",
    "cec": "Low",
    "ph_range": "7.5-8.5"
   }
  }
 ],
 "rules": [
  {
   "id": "R-SANDY-1",
   "type": "diagnostic",
   "archetype": "ARC-SANDY",
   "when": {
    "all": [
     "F-GRITTY"
    ]
   },
   "concludes": "Sandy texture. Texture is not by itself degradation; confirm low organic matter before diagnosing
     degradation.",
   "min_evidence": 1,
   "status": "pending_review"
  },
  {
   "id": "R-SALINE-1",
   "type": "diagnostic",
   "archetype": "ARC-SALINE",
   "when": {
    "any": [
     "F-SALT-CRUST",
     "F-SALT-SMELL"
    ],
    "also_any": [
     "F-STUNTED-BROWN",
     "F-SEA-NEAR"
    ]
   },
   "concludes": "Probable salinity. Salinity can only be confirmed by an EC measurement.",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-WATER-SAFETY",
   "legacy_name": "R14",
   "type": "safety_filter",
   "when": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect_data": {
    "require_water": [
     "low",
     "low-medium"
    ]
   }
  },
  {
   "id": "R-HEAVY-FEEDER",
   "legacy_name": "R13",
   "type": "safety_filter",
   "when": {
    "deficiency_any": [
     "low organic matter",
     "nutrient depletion",
     "low nutrient retention"
    ]
   },
   "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "archetype_stress_any": [
     "low fertility"
    ]
   },
   "effect_data": {
    "exclude_feeder_class": [
     "heavy"
    ]
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```
   }
  },
  {
   "id": "R-READINESS-GATE",
   "legacy_name": "R9",
   "type": "readiness",
   "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
     counts as not passed.",
   "status": "pending_review"
  },
  {
   "id": "R-MIXED",
   "type": "combination",
   "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
   "status": "pending_review"
  },
  {
   "id": "R-NO-ESTIMATE-AS-FACT",
   "type": "integrity",
   "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
     labelled 'measured'.",
   "status": "pending_review"
  }
 ],
 "procedures": [
  {
   "id": "PROC-ORG-MATTER",
   "name": "Compost / well-rotted manure incorporation",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "targets": [
    "low organic matter",
    "low water retention",
    "nutrient depletion"
   ],
   "prerequisites": [],
   "materials": [
    "compost or well-rotted manure"
   ],
   "rate": 3,
   "unit": "kg per m2",
   "steps": [
    "Spread evenly over the plot.",
    "Incorporate into the topsoil.",
    "Water lightly if water is available."
   ],
   "verification": [
    "Crumb structure test"
   ],
   "stabilization_days": 7,
   "safety": [
    "Use fully rotted material only."
   ],
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-MULCH",
   "name": "Organic mulch cover",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-SALINE",
    "ARC-DISTURBED"
   ],
   "targets": [
    "moisture loss",
    "surface protection",
    "wind erosion"
   ],
   "prerequisites": [],
   "materials": [
    "straw, dry grass or crop residue"
   ],
   "rate": 5,
   "unit": "cm depth",
   "steps": [
    "Apply after amendments are incorporated.",
    "Keep mulch in place against wind (anchor or wet down)."
   ],
   "verification": [
    "Soil under mulch stays visibly moist"
   ],
   "stabilization_days": 0,
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
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 {
  "id": "PROC-BIOCHAR",
  "name": "Biochar incorporation",
  "applies_to": [
   "ARC-SANDY"
  ],
  "targets": [
   "low nutrient retention",
   "low water retention"
  ],
  "prerequisites": [
   "PROC-ORG-MATTER"
  ],
  "materials": [
   "biochar, pre-charged with compost"
  ],
  "rate": 0.5,
  "unit": "kg per m2",
  "steps": [
   "Mix with compost before applying."
  ],
  "verification": [],
  "stabilization_days": 7,
  "phase": "B",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
 {
  "id": "PROC-WINDBREAK",
  "name": "Windbreak and ground cover",
  "applies_to": [
   "ARC-ARID",
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "targets": [
   "wind erosion",
   "wind damage"
  ],
  "prerequisites": [],
  "materials": [
   "tall hardy plants, brush or woven barriers"
  ],
  "rate": "1 barrier row per 10 m of exposed plot width, across the prevailing wind",
  "unit": "",
  "steps": [
   "Place barriers across the prevailing wind direction.",
   "Cover bare patches before planting."
  ],
  "verification": [
   "Visible dust movement reduced"
  ],
  "stabilization_days": 0,
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-CASE"
 },
 {
  "id": "PROC-LEACH",
  "name": "Salinity leaching",
  "applies_to": [
   "ARC-SALINE"
  ],
  "targets": [
   "salinity"
  ],
  "prerequisites": [
   "Confirmed low-salt water source",
   "Confirmed drainage out of the root zone"
  ],
  "materials": [
   "low-salt water"
  ],
  "rate": 150,
  "unit": "mm of low-salt water per leaching event; repeat until crust is gone",
  "steps": [
   "Flush with low-salt water only if water and drainage are confirmed."
  ],
  "verification": [
   "EC measurement or crust gone"
  ],
  "stabilization_days": 7,
  "safety": [
   "Leaching without drainage raises the water table and brings salt back up."
  ],
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 }
],
"readiness_criteria": [
 {
  "id": "RC-CRUMB",
```

```
   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-MOIST",
   "name": "Moisture retention",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID"
   ],
   "check": "Soil stays damp below the surface after watering.",
   "mandatory": true,
   "if_failed": "Add organic matter and mulch."
  },
  {
   "id": "RC-DUST",
   "name": "Surface stability",
   "applies_to": [
    "ARC-ARID",
    "ARC-SANDY",
    "ARC-SALINE"
   ],
   "check": "Wind no longer lifts loose soil from the surface.",
   "mandatory": true,
   "if_failed": "Add windbreak or ground cover."
  },
  {
   "id": "RC-SALT",
   "name": "Salt crust cleared",
   "applies_to": [
    "ARC-SALINE"
   ],
   "check": "No visible crust and a measured EC reading is available.",
   "mandatory": true,
   "if_failed": "Provide an EC reading or continue salinity management."
  },
  {
   "id": "RC-STAB",
   "name": "Stabilization window",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Required waiting time after amendments has passed.",
   "mandatory": true,
   "window_days": 3,
   "if_failed": "Wait for the stabilization window to pass, then repeat."
  }
 ],
 "crops": [
  {
   "id": "CROP-PMILLET",
   "name": "Pearl millet",
   "scientific": "Pennisetum glaucum",
   "category": "Cereal / dryland grain",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-SALINE",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility",
    "some salinity"
   ],
   "avoid": [
    "waterlogging"
   ],
   "water": "low",
   "light": "full",
   "duration_days": [
    65,
    90
   ],
   "feeder_class": "light",
   "role": "Drought and low-fertility tolerant cereal.",
   "planting": {
    "depth": "2-3 cm",
    "spacing": "10-15 cm in row, 45-50 cm between rows",
```

```
   "method": "Direct sow into prepared soil"
  },
  "care": {
   "diseases": [
    "Downy mildew",
    "Ergot"
   ],
   "pests": [
    "Stem borer"
   ],
   "controls": [
    "Clean or resistant seed",
    "Remove infected plants early",
    "Neem spray"
   ],
   "fertilizer": "Low demand; compost at bed prep and a small nitrogen top-dress at tillering."
  },
  "harvest": {
   "indicators": [
    "Ear heads dry, grains hard"
   ],
   "method": "Cut ear heads and dry"
  },
  "status": "pending_review",
  "source": "SRC-PRD",
  "phase_days": {
   "P1": "3-5",
   "P2": "20-25",
   "P3": "25-35",
   "P4": "7-10"
  }
 },
 {
  "id": "CROP-BARLEY",
  "name": "Barley",
  "scientific": "Hordeum vulgare",
  "category": "Cereal",
  "soil_compat": [
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "tolerates": [
   "salinity",
   "drought"
  ],
  "avoid": [
   "waterlogging"
  ],
  "water": "low-medium",
  "light": "full",
  "duration_days": [
   90,
   120
  ],
  "feeder_class": "moderate",
  "role": "Salt-tolerant cereal.",
  "planting": {
   "depth": "3-5 cm",
   "spacing": "Drill in rows 20 cm apart",
   "method": "Drill or broadcast and cover"
  },
  "care": {
   "diseases": [
    "Powdery mildew",
    "Net blotch",
    "Rust"
   ],
   "pests": [
    "Aphids",
    "Armyworm"
   ],
   "controls": [
    "Resistant varieties",
    "Crop rotation",
    "Neem spray"
   ],
   "fertilizer": "Moderate nitrogen, split; compost at bed prep; avoid chloride-heavy fertilizers."
  },
  "harvest": {
   "indicators": [
    "Heads golden",
    "Grains hard"
   ],
   "method": "Cut when dry, thresh, store below 12% moisture"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
   "P1": "5-10",
   "P2": "30-40",
   "P3": "30-40",
   "P4": "10-14"
  }
```

```
  },
  {
   "id": "CROP-TOMATO",
   "name": "Tomato",
   "scientific": "Solanum lycopersicum",
   "category": "Vegetable",
   "soil_compat": [],
   "tolerates": [],
   "avoid": [
    "drought",
    "low fertility",
    "salinity"
   ],
   "water": "high",
   "light": "full",
   "duration_days": [
    90,
    120
   ],
   "feeder_class": "heavy",
   "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
   "planting": {
    "depth": "0.5-1 cm in nursery; transplant at 5 cm",
    "spacing": "45 x 60 cm",
    "method": "Raise seedlings, then transplant"
   },
   "care": {
    "diseases": [
     "Early blight",
     "Late blight",
     "Bacterial wilt"
    ],
    "pests": [
     "Whitefly",
     "Fruit borer"
    ],
    "controls": [
     "Neem spray",
     "Crop rotation",
     "Stake and prune for airflow"
    ],
    "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
   },
   "harvest": {
    "indicators": [
     "Fruit fully coloured and slightly firm"
    ],
    "method": "Hand-pick ripe fruit regularly"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
    "P1": "5-8",
    "P2": "25-30",
    "P3": "30-45",
    "P4": "7-14"
   }
  }
 ],
 "phases": {
  "journey": [
   {
    "id": "P1",
    "name": "Germination"
   },
   {
    "id": "P2",
    "name": "Vegetative Growth"
   },
   {
    "id": "P3",
    "name": "Reproductive & Canopy Development"
   },
   {
    "id": "P4",
    "name": "Pre-Harvest"
   }
  ],
  "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
     data is unavailable."
 },
 "missing_data_messages": {
  "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
  "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
  "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
  "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
  "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
 }
}
```

6. kb/case4_pasture.json

Full file contents


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{
 "meta": {
  "name": "TERRAFORM KB - case4_pasture",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ],
  "water_confirm_features": [
   "F-STANDING-WATER"
  ],
  "conflicts": [
   [
    "F-STANDING-WATER",
    "F-CRACKED"
   ],
   [
    "F-STANDING-WATER",
    "F-RAIN-RARE"
   ]
  ],
  "text_limit": 3000
 },
 "sources": [
  {
   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
  },
  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
  },
  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
 ],
 "features": [
  {
   "id": "F-HARD-PACKED",
   "label": "Hard-packed ground",
   "kind": "text",
   "lexicon": "hard[- ]packed|compact|hard ground|pressed[- ]down|rock[- ]hard"
  },
  {
   "id": "F-SURFACE-POOLING",
   "label": "Water pools on surface instead of soaking in",
   "kind": "text",
   "lexicon": "water (sits|pools)|pools on (the )?surface|doesn.t soak|not soak|poor infiltration"
  },
  {
   "id": "F-BARE-PATCHES",
   "label": "Bare patches",
   "kind": "visual",
   "lexicon": "bare (patch|dirt|spot)|worn (down )?to"
  },
  {
   "id": "F-THIN-GRASS",
   "label": "Thin, patchy, grazed grass",
   "kind": "visual",
   "lexicon": "thin|patchy|overgraz|grazed|grazing|pasture"
  }
 ],
 "toggles": [
  {
   "id": "T-WATER",
   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
  },
  {
   "id": "T-SLOPE",
   "label": "Sloped Terrain",
   "sets": {
    "slope": true
```

```
    }
   },
   {
    "id": "T-ROCKY",
    "label": "Rocky",
    "sets": {
     "rocky": true
    }
   },
   {
    "id": "T-SHADE",
    "label": "Shaded",
    "sets": {
     "shaded": true
    },
    "cancels_features": [
     "F-NO-CANOPY"
    ]
   },
   {
    "id": "T-RAIN",
    "label": "Recent Rain",
    "sets": {
     "recent_rain": true
    }
   },
   {
    "id": "T-WIND",
    "label": "Strong Wind",
    "sets": {
     "strong_wind": true
    }
   },
   {
    "id": "T-SALT-AIR",
    "label": "Salty/Coastal Air",
    "sets": {
     "salt_air": true
    },
    "adds_features": [
     "F-SALT-SMELL"
    ]
   }
  ],
  "archetypes": [
   {
    "id": "ARC-COMPACTED",
    "name": "Compacted and Depleted",
    "criteria": "Hard-packed ground, poor infiltration, bare or thin grazed vegetation, low organic matter.",
    "variants": [
     "uniform",
     "uneven wear"
    ],
    "core_gaps": [
     "compaction",
     "poor infiltration",
     "nutrient and organic matter depletion"
    ],
    "status": "pending_review",
    "source": "SRC-CASE",
    "stress_tags": [
     "low fertility"
    ],
    "typical_estimates": {
     "texture_class": "Compacted loam / clay loam",
     "drainage": "Poor infiltration",
     "organic_matter": "Low",
     "cec": "Moderate",
     "ph_range": "5.5-7.0"
    }
   }
  ],
  "rules": [
   {
    "id": "R-COMP-1",
    "type": "diagnostic",
    "archetype": "ARC-COMPACTED",
    "when": {
     "any": [
      "F-HARD-PACKED",
      "F-SURFACE-POOLING"
     ],
     "also_any": [
      "F-BARE-PATCHES",
      "F-THIN-GRASS"
     ]
    },
    "concludes": "Compaction with depletion from overuse",
    "min_evidence": 2,
    "status": "pending_review"
   },
   {
    "id": "R-WATER-SAFETY",
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```
    "legacy_name": "R14",
    "type": "safety_filter",
    "when": {
     "water_access": [
      "unknown",
      "none"
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    },
    "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
    "configurable": true,
    "status": "pending_review",
    "condition": {
     "water_access": [
      "unknown",
      "none"
     ]
    },
    "effect_data": {
     "require_water": [
      "low",
      "low-medium"
     ]
    }
   },
   {
    "id": "R-HEAVY-FEEDER",
    "legacy_name": "R13",
    "type": "safety_filter",
    "when": {
     "deficiency_any": [
      "low organic matter",
      "nutrient depletion",
      "low nutrient retention"
     ]
    },
    "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
    "configurable": true,
    "status": "pending_review",
    "condition": {
     "archetype_stress_any": [
      "low fertility"
     ]
    },
    "effect_data": {
     "exclude_feeder_class": [
      "heavy"
     ]
    }
   },
   {
    "id": "R-READINESS-GATE",
    "legacy_name": "R9",
    "type": "readiness",
    "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
      counts as not passed.",
    "status": "pending_review"
   },
   {
    "id": "R-MIXED",
    "type": "combination",
    "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
    "status": "pending_review"
   },
   {
    "id": "R-NO-ESTIMATE-AS-FACT",
    "type": "integrity",
    "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
      labelled 'measured'.",
    "status": "pending_review"
   }
  ],
  "procedures": [
   {
    "id": "PROC-ORG-MATTER",
    "name": "Compost / well-rotted manure incorporation",
    "applies_to": [
     "ARC-SANDY",
     "ARC-ARID",
     "ARC-COMPACTED",
     "ARC-DISTURBED"
    ],
    "targets": [
     "low organic matter",
     "low water retention",
     "nutrient depletion"
    ],
    "prerequisites": [],
    "materials": [
     "compost or well-rotted manure"
    ],
    "rate": 3,
    "unit": "kg per m2",
    "steps": [
```

```
    "Spread evenly over the plot.",
    "Incorporate into the topsoil.",
    "Water lightly if water is available."
   ],
   "verification": [
    "Crumb structure test"
   ],
   "stabilization_days": 7,
   "safety": [
    "Use fully rotted material only."
   ],
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-DECOMPACT",
   "name": "Decompaction and aeration",
   "applies_to": [
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "targets": [
    "compaction",
    "poor infiltration"
   ],
   "prerequisites": [
    "Soil not saturated"
   ],
   "materials": [
    "fork, broadfork or subsoiler"
   ],
   "rate": "Loosen to 25-30 cm depth",
   "unit": "",
   "steps": [
    "Loosen the soil when moderately moist, not wet.",
    "Follow immediately with organic matter."
   ],
   "verification": [
    "Infiltration check"
   ],
   "stabilization_days": 3,
   "phase": "A",
   "status": "pending_review",
   "source": "SRC-CASE"
  },
  {
   "id": "PROC-COVER-CROP",
   "name": "Cover crop / green manure",
   "applies_to": [
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-ARID",
    "ARC-SANDY"
   ],
   "targets": [
    "nutrient depletion",
    "bare soil",
    "organic matter"
   ],
   "prerequisites": [
    "PROC-DECOMPACT or PROC-ORG-MATTER where relevant"
   ],
   "materials": [
    "legume cover crop seed"
   ],
   "rate": 5,
   "unit": "g legume seed per m2",
   "steps": [
    "Sow, grow, then cut and leave or dig in."
   ],
   "verification": [
    "Soil cover established"
   ],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  }
 ],
 "readiness_criteria": [
  {
   "id": "RC-CRUMB",
   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
```

```
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-INFIL",
   "name": "Infiltration check",
   "applies_to": [
    "ARC-WATERLOGGED",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Water poured on the surface soaks in without pooling.",
   "mandatory": true,
   "if_failed": "Repeat drainage or decompaction."
  },
  {
   "id": "RC-STAB",
   "name": "Stabilization window",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Required waiting time after amendments has passed.",
   "mandatory": true,
   "window_days": 3,
   "if_failed": "Wait for the stabilization window to pass, then repeat."
  }
 ],
 "crops": [
  {
   "id": "CROP-COWPEA",
   "name": "Cowpea / Black-eyed bean",
   "scientific": "Vigna unguiculata",
   "category": "Legume",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility"
   ],
   "avoid": [
    "waterlogging",
    "salinity"
   ],
   "water": "low-medium",
   "light": "full",
   "duration_days": [
    60,
    75
   ],
   "feeder_class": "light",
   "role": "Pioneer crop; fixes nitrogen on poor land.",
   "planting": {
    "depth": "2-4 cm",
    "spacing": "10 cm in row, 40-50 cm between rows",
    "method": "Direct sow into moist, prepared soil"
   },
   "care": {
    "diseases": [
     "Fusarium wilt"
    ],
    "pests": [
     "Aphids",
     "Pod borers"
    ],
    "controls": [
     "Neem-based spray"
    ],
    "fertilizer": "Little nitrogen needed (fixes its own). Compost at bed prep; phosphorus if available."
   },
   "harvest": {
    "indicators": [
     "Pods dry and yellow-brown"
    ],
    "method": "Hand-pick pods as they dry"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "3-7",
    "P2": "20-25",
    "P3": "20-30",
    "P4": "7-10"
   }
  },
  {
   "id": "CROP-SORGHUM",
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 "name": "Sorghum",
 "scientific": "Sorghum bicolor",
 "category": "Cereal / dryland grain",
 "soil_compat": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "heat",
  "marginal soil",
  "some waterlogging",
  "moderate salinity"
 ],
 "avoid": [],
 "water": "low",
 "light": "full",
 "duration_days": [
  90,
  120
 ],
 "feeder_class": "moderate",
 "role": "Drought and marginal-soil tolerant staple grain.",
 "planting": {
  "depth": "2-4 cm",
  "spacing": "15-20 cm in row, 60-75 cm between rows",
  "method": "Direct sow into prepared soil"
 },
 "care": {
  "diseases": [
   "Grain mold",
   "Anthracnose"
  ],
  "pests": [
   "Shoot fly",
   "Stem borer"
  ],
  "controls": [
   "Neem-based spray",
   "Intercrop with cowpea to disrupt stem borer",
   "Seed treatment with Trichoderma",
   "Reflective ribbon against birds"
  ],
  "fertilizer": "Modest nitrogen and phosphorus; well-rotted compost at bed prep; light top-dress at knee-height
    stage."
 },
 "harvest": {
  "indicators": [
   "Grain hard, heads mature"
  ],
  "method": "Cut heads, dry before storage"
 },
 "status": "pending_review",
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
  "P3": "30-40",
  "P4": "10-14"
 }
},
{
 "id": "CROP-PIGEONPEA",
 "name": "Pigeon pea",
 "scientific": "Cajanus cajan",
 "category": "Legume",
 "soil_compat": [
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "compaction (deep roots)"
 ],
 "avoid": [
  "waterlogging"
 ],
 "water": "low",
 "light": "full",
 "duration_days": [
  150,
  180
 ],
 "feeder_class": "light",
 "role": "Deep-rooted legume that helps recover compacted, depleted soil.",
 "planting": {
  "depth": "3-5 cm",
  "spacing": "30 cm in row, 75-90 cm between rows",
  "method": "Direct sow into loosened soil"
 },
 "care": {
```

```
    "diseases": [
     "Fusarium wilt",
     "Sterility mosaic",
     "Phytophthora blight"
    ],
    "pests": [
     "Pod borer",
     "Pod fly",
     "Aphids"
    ],
    "controls": [
     "Neem spray",
     "Intercrop with cereals",
     "Resistant varieties",
     "Pheromone traps"
    ],
    "fertilizer": "Little nitrogen; phosphorus helpful; compost at bed prep."
   },
   "harvest": {
    "indicators": [
     "Pods brown and dry",
     "Seeds hard"
    ],
    "method": "Hand-pick or cut dry pods in passes; thresh"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
    "P1": "5-10",
    "P2": "40-60",
    "P3": "50-70",
    "P4": "10-15"
   }
  },
  {
   "id": "CROP-TOMATO",
   "name": "Tomato",
   "scientific": "Solanum lycopersicum",
   "category": "Vegetable",
   "soil_compat": [],
   "tolerates": [],
   "avoid": [
    "drought",
    "low fertility",
    "salinity"
   ],
   "water": "high",
   "light": "full",
   "duration_days": [
    90,
    120
   ],
   "feeder_class": "heavy",
   "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
   "planting": {
    "depth": "0.5-1 cm in nursery; transplant at 5 cm",
    "spacing": "45 x 60 cm",
    "method": "Raise seedlings, then transplant"
   },
   "care": {
    "diseases": [
     "Early blight",
     "Late blight",
     "Bacterial wilt"
    ],
    "pests": [
     "Whitefly",
     "Fruit borer"
    ],
    "controls": [
     "Neem spray",
     "Crop rotation",
     "Stake and prune for airflow"
    ],
    "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
   },
   "harvest": {
    "indicators": [
     "Fruit fully coloured and slightly firm"
    ],
    "method": "Hand-pick ripe fruit regularly"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
    "P1": "5-8",
    "P2": "25-30",
    "P3": "30-45",
    "P4": "7-14"
   }
  }
 ],
 "phases": {
```

```
  "journey": [
   {
    "id": "P1",
    "name": "Germination"
   },
   {
    "id": "P2",
    "name": "Vegetative Growth"
   },
   {
    "id": "P3",
    "name": "Reproductive & Canopy Development"
   },
   {
    "id": "P4",
    "name": "Pre-Harvest"
   }
  ],
  "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
     data is unavailable."
 },
 "missing_data_messages": {
  "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
  "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
  "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
  "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
  "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
 }
}
```

7. kb/case5_deforested.json

Full file contents


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{
 "meta": {
  "name": "TERRAFORM KB - case5_deforested",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ],
  "water_confirm_features": [
   "F-STANDING-WATER"
  ],
  "conflicts": [
   [
    "F-STANDING-WATER",
    "F-CRACKED"
   ],
   [
    "F-STANDING-WATER",
    "F-RAIN-RARE"
   ]
  ],
  "text_limit": 3000
 },
 "sources": [
  {
   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
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  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
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  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
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 "features": [
  {
   "id": "F-HARD-PACKED",
   "label": "Hard-packed ground",
   "kind": "text",
   "lexicon": "hard[- ]packed|compact|hard ground|pressed[- ]down|rock[- ]hard"
  },
  {
   "id": "F-STUMPS",
   "label": "Stumps, debris or vehicle tracks",
   "kind": "visual",
   "lexicon": "stump|debris|tracks|cleared|deforest|felled|logging"
  },
  {
   "id": "F-NO-CANOPY",
   "label": "No canopy or shade",
   "kind": "visual",
   "lexicon": "no (canopy|shade|trees)|exposed to (the )?sun|full sun|open to the sun|cleared"
  }
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 "toggles": [
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   "id": "T-WATER",
   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
  },
  {
   "id": "T-SLOPE",
   "label": "Sloped Terrain",
   "sets": {
    "slope": true
   }
  },
  {
   "id": "T-ROCKY",
   "label": "Rocky",
   "sets": {
```

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    "rocky": true
   }
  },
  {
   "id": "T-SHADE",
   "label": "Shaded",
   "sets": {
    "shaded": true
   },
   "cancels_features": [
    "F-NO-CANOPY"
   ]
  },
  {
   "id": "T-RAIN",
   "label": "Recent Rain",
   "sets": {
    "recent_rain": true
   }
  },
  {
   "id": "T-WIND",
   "label": "Strong Wind",
   "sets": {
    "strong_wind": true
   }
  },
  {
   "id": "T-SALT-AIR",
   "label": "Salty/Coastal Air",
   "sets": {
    "salt_air": true
   },
   "adds_features": [
    "F-SALT-SMELL"
   ]
  }
 ],
 "archetypes": [
  {
   "id": "ARC-DISTURBED",
   "name": "Disturbed, Compacted, Sun-Exposed",
   "criteria": "Cleared land with tracks, stumps or debris; no canopy; compacted.",
   "core_gaps": [
    "compaction",
    "heat and sun exposure",
    "no surface cover"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "stress_tags": [
    "heat",
    "low fertility"
   ],
   "typical_estimates": {
    "texture_class": "Compacted loam / clay loam",
    "drainage": "Poor infiltration",
    "organic_matter": "Low (topsoil lost)",
    "cec": "Low to moderate",
    "ph_range": "5.5-7.0"
   }
  }
 ],
 "rules": [
  {
   "id": "R-DIST-1",
   "type": "diagnostic",
   "archetype": "ARC-DISTURBED",
   "when": {
    "any": [
     "F-STUMPS"
    ],
    "also_any": [
     "F-NO-CANOPY",
     "F-HARD-PACKED"
    ]
   },
   "concludes": "Disturbed, compacted land with sun exposure as a separate stressor",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-WATER-SAFETY",
   "legacy_name": "R14",
   "type": "safety_filter",
   "when": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
   "configurable": true,
```

```
   "status": "pending_review",
   "condition": {
    "water_access": [
     "unknown",
     "none"
    ]
   },
   "effect_data": {
    "require_water": [
     "low",
     "low-medium"
    ]
   }
  },
  {
   "id": "R-HEAVY-FEEDER",
   "legacy_name": "R13",
   "type": "safety_filter",
   "when": {
    "deficiency_any": [
     "low organic matter",
     "nutrient depletion",
     "low nutrient retention"
    ]
   },
   "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
   "configurable": true,
   "status": "pending_review",
   "condition": {
    "archetype_stress_any": [
     "low fertility"
    ]
   },
   "effect_data": {
    "exclude_feeder_class": [
     "heavy"
    ]
   }
  },
  {
   "id": "R-READINESS-GATE",
   "legacy_name": "R9",
   "type": "readiness",
   "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
     counts as not passed.",
   "status": "pending_review"
  },
  {
   "id": "R-MIXED",
   "type": "combination",
   "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
   "status": "pending_review"
  },
  {
   "id": "R-NO-ESTIMATE-AS-FACT",
   "type": "integrity",
   "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
     labelled 'measured'.",
   "status": "pending_review"
  }
 ],
 "procedures": [
  {
   "id": "PROC-ORG-MATTER",
   "name": "Compost / well-rotted manure incorporation",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "targets": [
    "low organic matter",
    "low water retention",
    "nutrient depletion"
   ],
   "prerequisites": [],
   "materials": [
    "compost or well-rotted manure"
   ],
   "rate": 3,
   "unit": "kg per m2",
   "steps": [
    "Spread evenly over the plot.",
    "Incorporate into the topsoil.",
    "Water lightly if water is available."
   ],
   "verification": [
    "Crumb structure test"
   ],
   "stabilization_days": 7,
   "safety": [
    "Use fully rotted material only."
```

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 "phase": "A",
 "status": "pending_review",
 "source": "SRC-AGRI-GEN"
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 "id": "PROC-MULCH",
 "name": "Organic mulch cover",
 "applies_to": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-SALINE",
  "ARC-DISTURBED"
 ],
 "targets": [
  "moisture loss",
  "surface protection",
  "wind erosion"
 ],
 "prerequisites": [],
 "materials": [
  "straw, dry grass or crop residue"
 ],
 "rate": 5,
 "unit": "cm depth",
 "steps": [
  "Apply after amendments are incorporated.",
  "Keep mulch in place against wind (anchor or wet down)."
 ],
 "verification": [
  "Soil under mulch stays visibly moist"
 ],
 "stabilization_days": 0,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-AGRI-GEN"
},
{
 "id": "PROC-DECOMPACT",
 "name": "Decompaction and aeration",
 "applies_to": [
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "targets": [
  "compaction",
  "poor infiltration"
 ],
 "prerequisites": [
  "Soil not saturated"
 ],
 "materials": [
  "fork, broadfork or subsoiler"
 ],
 "rate": "Loosen to 25-30 cm depth",
 "unit": "",
 "steps": [
  "Loosen the soil when moderately moist, not wet.",
  "Follow immediately with organic matter."
 ],
 "verification": [
  "Infiltration check"
 ],
 "stabilization_days": 3,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-CASE"
},
{
 "id": "PROC-COVER-CROP",
 "name": "Cover crop / green manure",
 "applies_to": [
  "ARC-COMPACTED",
  "ARC-DISTURBED",
  "ARC-ARID",
  "ARC-SANDY"
 ],
 "targets": [
  "nutrient depletion",
  "bare soil",
  "organic matter"
 ],
 "prerequisites": [
  "PROC-DECOMPACT or PROC-ORG-MATTER where relevant"
 ],
 "materials": [
  "legume cover crop seed"
 ],
 "rate": 5,
 "unit": "g legume seed per m2",
 "steps": [
  "Sow, grow, then cut and leave or dig in."
 ],
```

```
   "verification": [
    "Soil cover established"
   ],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-SHADE",
   "name": "Temporary shade or surface protection",
   "applies_to": [
    "ARC-DISTURBED",
    "ARC-ARID"
   ],
   "targets": [
    "heat exposure"
   ],
   "prerequisites": [],
   "materials": [
    "shade cloth, mulch or nurse plants"
   ],
   "rate": "30-50% shade cloth or mulch cover over seedlings until established",
   "unit": "",
   "steps": [
    "Protect seedlings during establishment."
   ],
   "verification": [],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-CASE"
  }
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 "readiness_criteria": [
  {
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   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-INFIL",
   "name": "Infiltration check",
   "applies_to": [
    "ARC-WATERLOGGED",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Water poured on the surface soaks in without pooling.",
   "mandatory": true,
   "if_failed": "Repeat drainage or decompaction."
  },
  {
   "id": "RC-STAB",
   "name": "Stabilization window",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Required waiting time after amendments has passed.",
   "mandatory": true,
   "window_days": 3,
   "if_failed": "Wait for the stabilization window to pass, then repeat."
  }
 ],
 "crops": [
  {
   "id": "CROP-COWPEA",
   "name": "Cowpea / Black-eyed bean",
   "scientific": "Vigna unguiculata",
   "category": "Legume",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility"
   ],
```

```
  "avoid": [
   "waterlogging",
   "salinity"
  ],
  "water": "low-medium",
  "light": "full",
  "duration_days": [
   60,
   75
  ],
  "feeder_class": "light",
  "role": "Pioneer crop; fixes nitrogen on poor land.",
  "planting": {
   "depth": "2-4 cm",
   "spacing": "10 cm in row, 40-50 cm between rows",
   "method": "Direct sow into moist, prepared soil"
  },
  "care": {
   "diseases": [
    "Fusarium wilt"
   ],
   "pests": [
    "Aphids",
    "Pod borers"
   ],
   "controls": [
    "Neem-based spray"
   ],
   "fertilizer": "Little nitrogen needed (fixes its own). Compost at bed prep; phosphorus if available."
  },
  "harvest": {
   "indicators": [
    "Pods dry and yellow-brown"
   ],
   "method": "Hand-pick pods as they dry"
  },
  "status": "pending_review",
  "source": "SRC-PRD",
  "phase_days": {
   "P1": "3-7",
   "P2": "20-25",
   "P3": "20-30",
   "P4": "7-10"
  }
 },
 {
  "id": "CROP-SORGHUM",
  "name": "Sorghum",
  "scientific": "Sorghum bicolor",
  "category": "Cereal / dryland grain",
  "soil_compat": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-COMPACTED",
   "ARC-DISTURBED"
  ],
  "tolerates": [
   "drought",
   "heat",
   "marginal soil",
   "some waterlogging",
   "moderate salinity"
  ],
  "avoid": [],
  "water": "low",
  "light": "full",
  "duration_days": [
   90,
   120
  ],
  "feeder_class": "moderate",
  "role": "Drought and marginal-soil tolerant staple grain.",
  "planting": {
   "depth": "2-4 cm",
   "spacing": "15-20 cm in row, 60-75 cm between rows",
   "method": "Direct sow into prepared soil"
  },
  "care": {
   "diseases": [
    "Grain mold",
    "Anthracnose"
   ],
   "pests": [
    "Shoot fly",
    "Stem borer"
   ],
   "controls": [
    "Neem-based spray",
    "Intercrop with cowpea to disrupt stem borer",
    "Seed treatment with Trichoderma",
    "Reflective ribbon against birds"
   ],
   "fertilizer": "Modest nitrogen and phosphorus; well-rotted compost at bed prep; light top-dress at knee-height
```

```
    stage."
   },
   "harvest": {
    "indicators": [
     "Grain hard, heads mature"
    ],
    "method": "Cut heads, dry before storage"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "5-10",
    "P2": "30-40",
    "P3": "30-40",
    "P4": "10-14"
   }
  },
  {
   "id": "CROP-PMILLET",
   "name": "Pearl millet",
   "scientific": "Pennisetum glaucum",
   "category": "Cereal / dryland grain",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-SALINE",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat",
    "low fertility",
    "some salinity"
   ],
   "avoid": [
    "waterlogging"
   ],
   "water": "low",
   "light": "full",
   "duration_days": [
    65,
    90
   ],
   "feeder_class": "light",
   "role": "Drought and low-fertility tolerant cereal.",
   "planting": {
    "depth": "2-3 cm",
    "spacing": "10-15 cm in row, 45-50 cm between rows",
    "method": "Direct sow into prepared soil"
   },
   "care": {
    "diseases": [
     "Downy mildew",
     "Ergot"
    ],
    "pests": [
     "Stem borer"
    ],
    "controls": [
     "Clean or resistant seed",
     "Remove infected plants early",
     "Neem spray"
    ],
    "fertilizer": "Low demand; compost at bed prep and a small nitrogen top-dress at tillering."
   },
   "harvest": {
    "indicators": [
     "Ear heads dry, grains hard"
    ],
    "method": "Cut ear heads and dry"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "3-5",
    "P2": "20-25",
    "P3": "25-35",
    "P4": "7-10"
   }
  },
  {
   "id": "CROP-AMARANTH",
   "name": "Amaranth",
   "scientific": "Amaranthus spp.",
   "category": "Grain / leafy dual-purpose",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat"
```

```
 ],
 "avoid": [],
 "water": "low-medium",
 "light": "full",
 "duration_days": [
  30,
  40
 ],
 "duration_note": "Leafy harvest 30-40 days; grain harvest 90-110 days.",
 "feeder_class": "light",
 "role": "Pioneer crop where soil test results are uncertain.",
 "planting": {
  "depth": "0.5-1 cm (tiny seed; mix with sand)",
  "spacing": "10-15 cm in row, 30-45 cm between rows; thin seedlings",
  "method": "Broadcast or drill shallowly into fine, moist soil"
 },
 "care": {
  "diseases": [
   "Damping-off",
   "Leaf spot",
   "Root rot in wet soil"
  ],
  "pests": [
   "Aphids"
  ],
  "controls": [
   "Neem spray",
   "Avoid overwatering",
   "Thin plants for airflow"
  ],
  "fertilizer": "Light feeder; compost at bed prep and a light nitrogen feed after the first leaf cut."
 },
 "harvest": {
  "indicators": [
   "Leaves tender (leafy)",
   "Seed heads dry (grain)"
  ],
  "method": "Cut leaves early or seed heads late"
 },
 "status": "pending_review",
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "3-5",
  "P2": "15-25",
  "P3": "40-60 (grain)",
  "P4": "7-10"
 }
},
{
 "id": "CROP-PIGEONPEA",
 "name": "Pigeon pea",
 "scientific": "Cajanus cajan",
 "category": "Legume",
 "soil_compat": [
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "compaction (deep roots)"
 ],
 "avoid": [
  "waterlogging"
 ],
 "water": "low",
 "light": "full",
 "duration_days": [
  150,
  180
 ],
 "feeder_class": "light",
 "role": "Deep-rooted legume that helps recover compacted, depleted soil.",
 "planting": {
  "depth": "3-5 cm",
  "spacing": "30 cm in row, 75-90 cm between rows",
  "method": "Direct sow into loosened soil"
 },
 "care": {
  "diseases": [
   "Fusarium wilt",
   "Sterility mosaic",
   "Phytophthora blight"
  ],
  "pests": [
   "Pod borer",
   "Pod fly",
   "Aphids"
  ],
  "controls": [
   "Neem spray",
   "Intercrop with cereals",
   "Resistant varieties",
   "Pheromone traps"
```

```
    ],
    "fertilizer": "Little nitrogen; phosphorus helpful; compost at bed prep."
   },
   "harvest": {
    "indicators": [
     "Pods brown and dry",
     "Seeds hard"
    ],
    "method": "Hand-pick or cut dry pods in passes; thresh"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
    "P1": "5-10",
    "P2": "40-60",
    "P3": "50-70",
    "P4": "10-15"
   }
  },
  {
   "id": "CROP-TOMATO",
   "name": "Tomato",
   "scientific": "Solanum lycopersicum",
   "category": "Vegetable",
   "soil_compat": [],
   "tolerates": [],
   "avoid": [
    "drought",
    "low fertility",
    "salinity"
   ],
   "water": "high",
   "light": "full",
   "duration_days": [
    90,
    120
   ],
   "feeder_class": "heavy",
   "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
   "planting": {
    "depth": "0.5-1 cm in nursery; transplant at 5 cm",
    "spacing": "45 x 60 cm",
    "method": "Raise seedlings, then transplant"
   },
   "care": {
    "diseases": [
     "Early blight",
     "Late blight",
     "Bacterial wilt"
    ],
    "pests": [
     "Whitefly",
     "Fruit borer"
    ],
    "controls": [
     "Neem spray",
     "Crop rotation",
     "Stake and prune for airflow"
    ],
    "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
   },
   "harvest": {
    "indicators": [
     "Fruit fully coloured and slightly firm"
    ],
    "method": "Hand-pick ripe fruit regularly"
   },
   "status": "pending_review",
   "source": "SRC-AGRI-GEN",
   "phase_days": {
    "P1": "5-8",
    "P2": "25-30",
    "P3": "30-45",
    "P4": "7-14"
   }
  }
 ],
 "phases": {
  "journey": [
   {
    "id": "P1",
    "name": "Germination"
   },
   {
    "id": "P2",
    "name": "Vegetative Growth"
   },
   {
    "id": "P3",
    "name": "Reproductive & Canopy Development"
   },
   {
    "id": "P4",
```

```
    "name": "Pre-Harvest"
   }
  ],
  "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
     data is unavailable."
 },
 "missing_data_messages": {
  "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
  "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
  "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
  "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
  "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
 }
}
```

8. knowledge_base.json

Full file contents


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{
 "meta": {
  "name": "TERRAFORM Knowledge Base",
  "kb_version": "0.1.0-draft",
  "updated": "2026-10-13",
  "note": "Seed data. Every record is pending_review until an administrator approves it. Only approved records may be
    shown as verified recommendations. A null value means 'no verified data' and the system must show a missing-data
    message instead of inventing a number.",
  "validation_statuses": [
   "approved",
   "pending_review",
   "deprecated"
  ],
  "readiness_states": [
   "NOT_READY",
   "CONDITIONALLY_READY",
   "READY"
  ]
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 "sources": [
  {
   "id": "SRC-CASE",
   "title": "TERRAFORM Team Case Studies (AI Club, Sep 2026)",
   "type": "internal"
  },
  {
   "id": "SRC-PRD",
   "title": "TERRAFORM PRD",
   "type": "internal"
  },
  {
   "id": "SRC-AGRI-GEN",
   "title": "General agronomy references (to be cited by reviewer before approval)",
   "type": "pending_citation"
  }
 ],
 "features": [
  {
   "id": "F-CRACKED",
   "label": "Cracked soil surface",
   "kind": "visual"
  },
  {
   "id": "F-LOOSE-DUST",
   "label": "Loose soil or dust lifted by wind",
   "kind": "visual"
  },
  {
   "id": "F-GRITTY",
   "label": "Gritty, loose, sandy texture",
   "kind": "text"
  },
  {
   "id": "F-NO-VEG",
   "label": "No or very sparse vegetation",
   "kind": "visual"
  },
  {
   "id": "F-STANDING-WATER",
   "label": "Standing water or puddles",
   "kind": "visual"
  },
  {
   "id": "F-SOUR-SMELL",
   "label": "Sour or musty soil smell",
   "kind": "text"
  },
  {
   "id": "F-YELLOW-WEEDS",
   "label": "Yellowing weeds with soft or rotting roots",
   "kind": "visual"
  },
  {
   "id": "F-SLOW-DRAIN",
   "label": "Drains slowly, stays wet after rain",
   "kind": "text"
  },
  {
   "id": "F-DARK-STICKY",
   "label": "Dark, sticky clay",
   "kind": "text"
  },
  {
   "id": "F-SALT-CRUST",
   "label": "White or pale crust on surface",
   "kind": "visual"
  },
  {
   "id": "F-SALT-SMELL",
```

```
   "label": "Salty air or smell",
   "kind": "text"
  },
  {
   "id": "F-STUNTED-BROWN",
   "label": "Stunted, browning vegetation",
   "kind": "visual"
  },
  {
   "id": "F-SEA-NEAR",
   "label": "Sea or shoreline nearby",
   "kind": "text"
  },
  {
   "id": "F-HARD-PACKED",
   "label": "Hard-packed ground",
   "kind": "text"
  },
  {
   "id": "F-SURFACE-POOLING",
   "label": "Water pools on surface instead of soaking in",
   "kind": "text"
  },
  {
   "id": "F-BARE-PATCHES",
   "label": "Bare patches",
   "kind": "visual"
  },
  {
   "id": "F-THIN-GRASS",
   "label": "Thin, patchy, grazed grass",
   "kind": "visual"
  },
  {
   "id": "F-STUMPS",
   "label": "Stumps, debris or vehicle tracks",
   "kind": "visual"
  },
  {
   "id": "F-NO-CANOPY",
   "label": "No canopy or shade",
   "kind": "visual"
  },
  {
   "id": "F-HIGH-HEAT",
   "label": "Daytime temperatures reported above 40 C",
   "kind": "text"
  },
  {
   "id": "F-RAIN-RARE",
   "label": "Rainfall rare or unreliable",
   "kind": "text"
  }
 ],
 "toggles": [
  {
   "id": "T-WATER",
   "label": "Water Source Nearby",
   "sets": {
    "water_access": "confirmed"
   }
  },
  {
   "id": "T-SLOPE",
   "label": "Sloped Terrain",
   "sets": {
    "slope": true
   }
  },
  {
   "id": "T-ROCKY",
   "label": "Rocky",
   "sets": {
    "rocky": true
   }
  },
  {
   "id": "T-SHADE",
   "label": "Shaded",
   "sets": {
    "shaded": true
   }
  },
  {
   "id": "T-RAIN",
   "label": "Recent Rain",
   "sets": {
    "recent_rain": true
   }
  },
  {
   "id": "T-WIND",
   "label": "Strong Wind",
```

```
   "sets": {
    "strong_wind": true
   }
  },
  {
   "id": "T-SALT-AIR",
   "label": "Salty/Coastal Air",
   "sets": {
    "salt_air": true
   }
  }
 ],
 "archetypes": [
  {
   "id": "ARC-SANDY",
   "name": "Sandy / Loamy Sand (Degraded)",
   "criteria": "Gritty, loose texture; low cohesion; rapid drainage.",
   "physical_profile": "Loose, particulate, low aggregate stability",
   "core_gaps": [
    "low water-holding capacity",
    "low organic matter",
    "low nutrient retention"
   ],
   "typical_estimates": {
    "texture_class": "Sand / loamy sand",
    "drainage": "Excessive",
    "ph_range": "5.5-7.0",
    "organic_matter": "Low",
    "cec": "Low"
   },
   "estimate_label": "estimate, not laboratory measurement",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "ARC-ARID",
   "name": "Arid / Drought-Stressed",
   "criteria": "Dry, low organic matter, poor moisture retention, unreliable rain.",
   "variants": [
    "cracked-compacted (retention dominant)",
    "loose-eroding (wind erosion dominant)"
   ],
   "core_gaps": [
    "water scarcity",
    "wind erosion",
    "poor structure"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "typical_estimates": {
    "texture_class": "Clay-loam crust to loose sand",
    "drainage": "Poor when crusted, rapid when sandy",
    "organic_matter": "Low",
    "cec": "Low to moderate",
    "ph_range": "7.0-8.5"
   }
  },
  {
   "id": "ARC-WATERLOGGED",
   "name": "Waterlogged / Poorly Drained",
   "criteria": "Standing water or slow drainage; sour smell; oxygen-starved root zone.",
   "variants": [
    "recently flooded (days)",
    "prolonged waterlogging",
    "heavy slow-draining without visible pooling"
   ],
   "core_gaps": [
    "drainage",
    "root-zone oxygen",
    "nutrient loss",
    "disease risk"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "typical_estimates": {
    "texture_class": "Clay / silty clay",
    "drainage": "Very poor",
    "organic_matter": "Moderate, partly decomposed",
    "cec": "Moderate to high",
    "ph_range": "5.5-6.5"
   }
  },
  {
   "id": "ARC-SALINE",
   "name": "Saline / Salt-Affected",
   "criteria": "Salt crust or salty air, stunted browning plants, coastal context.",
   "variants": [
    "visible crust",
    "no crust but salty smell and stunting",
    "salt spray without saline soil"
   ],
   "core_gaps": [
    "salinity",
```

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    "wind exposure",
    "low water retention if sandy"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "typical_estimates": {
    "texture_class": "Sand / loamy sand",
    "drainage": "Rapid",
    "organic_matter": "Low",
    "cec": "Low",
    "ph_range": "7.5-8.5"
   }
  },
  {
   "id": "ARC-COMPACTED",
   "name": "Compacted and Depleted",
   "criteria": "Hard-packed ground, poor infiltration, bare or thin grazed vegetation, low organic matter.",
   "variants": [
    "uniform",
    "uneven wear"
   ],
   "core_gaps": [
    "compaction",
    "poor infiltration",
    "nutrient and organic matter depletion"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "typical_estimates": {
    "texture_class": "Compacted loam / clay loam",
    "drainage": "Poor infiltration",
    "organic_matter": "Low",
    "cec": "Moderate",
    "ph_range": "5.5-7.0"
   }
  },
  {
   "id": "ARC-DISTURBED",
   "name": "Disturbed, Compacted, Sun-Exposed",
   "criteria": "Cleared land with tracks, stumps or debris; no canopy; compacted.",
   "core_gaps": [
    "compaction",
    "heat and sun exposure",
    "no surface cover"
   ],
   "status": "pending_review",
   "source": "SRC-CASE",
   "typical_estimates": {
    "texture_class": "Compacted loam / clay loam",
    "drainage": "Poor infiltration",
    "organic_matter": "Low (topsoil lost)",
    "cec": "Low to moderate",
    "ph_range": "5.5-7.0"
   }
  }
 ],
 "rules": [
  {
   "id": "R-ARID-1",
   "type": "diagnostic",
   "archetype": "ARC-ARID",
   "when": {
    "all": [
     "F-NO-VEG"
    ],
    "any": [
     "F-CRACKED",
     "F-RAIN-RARE",
     "F-HIGH-HEAT"
    ]
   },
   "concludes": "Drought stress with poor moisture retention",
   "min_evidence": 2,
   "status": "pending_review"
  },
  {
   "id": "R-ARID-2",
   "type": "diagnostic",
   "archetype": "ARC-ARID",
   "when": {
    "all": [
     "F-LOOSE-DUST"
    ],
    "any": [
     "F-NO-VEG",
     "F-GRITTY",
     "F-CRACKED"
    ]
   },
   "concludes": "Active wind erosion in addition to dryness",
   "min_evidence": 2,
   "status": "pending_review"
  },
```

```
{
 "id": "R-SANDY-1",
 "type": "diagnostic",
 "archetype": "ARC-SANDY",
 "when": {
  "all": [
   "F-GRITTY"
  ]
 },
 "concludes": "Sandy texture. Texture is not by itself degradation; confirm low organic matter before diagnosing
   degradation.",
 "min_evidence": 1,
 "status": "pending_review"
},
{
 "id": "R-WET-1",
 "type": "diagnostic",
 "archetype": "ARC-WATERLOGGED",
 "when": {
  "any": [
   "F-STANDING-WATER",
   "F-SLOW-DRAIN"
  ],
  "also_any": [
   "F-SOUR-SMELL",
   "F-YELLOW-WEEDS",
   "F-DARK-STICKY"
  ]
 },
 "concludes": "Waterlogging with root-zone oxygen stress",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-SALINE-1",
 "type": "diagnostic",
 "archetype": "ARC-SALINE",
 "when": {
  "any": [
   "F-SALT-CRUST",
   "F-SALT-SMELL"
  ],
  "also_any": [
   "F-STUNTED-BROWN",
   "F-SEA-NEAR"
  ]
 },
 "concludes": "Probable salinity. Salinity can only be confirmed by an EC measurement.",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-COMP-1",
 "type": "diagnostic",
 "archetype": "ARC-COMPACTED",
 "when": {
  "any": [
   "F-HARD-PACKED",
   "F-SURFACE-POOLING"
  ],
  "also_any": [
   "F-BARE-PATCHES",
   "F-THIN-GRASS"
  ]
 },
 "concludes": "Compaction with depletion from overuse",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-DIST-1",
 "type": "diagnostic",
 "archetype": "ARC-DISTURBED",
 "when": {
  "any": [
   "F-STUMPS"
  ],
  "also_any": [
   "F-NO-CANOPY",
   "F-HARD-PACKED"
  ]
 },
 "concludes": "Disturbed, compacted land with sun exposure as a separate stressor",
 "min_evidence": 2,
 "status": "pending_review"
},
{
 "id": "R-WATER-SAFETY",
 "legacy_name": "R14",
 "type": "safety_filter",
 "when": {
  "water_access": [
   "unknown",
```

```
      "none"
     ]
    },
    "effect": "Restrict crops to those with water_requirement 'low' or 'low-medium'.",
    "configurable": true,
    "status": "pending_review"
   },
   {
    "id": "R-HEAVY-FEEDER",
    "legacy_name": "R13",
    "type": "safety_filter",
    "when": {
     "deficiency_any": [
      "low organic matter",
      "nutrient depletion",
      "low nutrient retention"
     ]
    },
    "effect": "Exclude crops with feeder_class 'heavy' (for example tomato, cucumber).",
    "configurable": true,
    "status": "pending_review"
   },
   {
    "id": "R-READINESS-GATE",
    "legacy_name": "R9",
    "type": "readiness",
    "effect": "Planting is blocked until the readiness criteria for the diagnosed archetype are met. A missing result
      counts as not passed.",
    "status": "pending_review"
   },
   {
    "id": "R-MIXED",
    "type": "combination",
    "effect": "More than one archetype may be returned when each meets its min_evidence. Never force a single label.",
    "status": "pending_review"
   },
   {
    "id": "R-NO-ESTIMATE-AS-FACT",
    "type": "integrity",
    "effect": "Properties inferred from text or image are labelled 'estimate'. Only user-supplied lab values may be
      labelled 'measured'.",
    "status": "pending_review"
   }
  ],
  "procedures": [
   {
    "id": "PROC-ORG-MATTER",
    "name": "Compost / well-rotted manure incorporation",
    "applies_to": [
     "ARC-SANDY",
     "ARC-ARID",
     "ARC-COMPACTED",
     "ARC-DISTURBED"
    ],
    "targets": [
     "low organic matter",
     "low water retention",
     "nutrient depletion"
    ],
    "prerequisites": [],
    "materials": [
     "compost or well-rotted manure"
    ],
    "rate": 3,
    "unit": "kg per m2",
    "steps": [
     "Spread evenly over the plot.",
     "Incorporate into the topsoil.",
     "Water lightly if water is available."
    ],
    "verification": [
     "Crumb structure test"
    ],
    "stabilization_days": 7,
    "safety": [
     "Use fully rotted material only."
    ],
    "phase": "A",
    "status": "pending_review",
    "source": "SRC-AGRI-GEN"
   },
   {
    "id": "PROC-MULCH",
    "name": "Organic mulch cover",
    "applies_to": [
     "ARC-SANDY",
     "ARC-ARID",
     "ARC-SALINE",
     "ARC-DISTURBED"
    ],
    "targets": [
     "moisture loss",
     "surface protection",
```

```
   "wind erosion"
  ],
  "prerequisites": [],
  "materials": [
   "straw, dry grass or crop residue"
  ],
  "rate": 5,
  "unit": "cm depth",
  "steps": [
   "Apply after amendments are incorporated.",
   "Keep mulch in place against wind (anchor or wet down)."
  ],
  "verification": [
   "Soil under mulch stays visibly moist"
  ],
  "stabilization_days": 0,
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
 {
  "id": "PROC-BIOCHAR",
  "name": "Biochar incorporation",
  "applies_to": [
   "ARC-SANDY"
  ],
  "targets": [
   "low nutrient retention",
   "low water retention"
  ],
  "prerequisites": [
   "PROC-ORG-MATTER"
  ],
  "materials": [
   "biochar, pre-charged with compost"
  ],
  "rate": 0.5,
  "unit": "kg per m2",
  "steps": [
   "Mix with compost before applying."
  ],
  "verification": [],
  "stabilization_days": 7,
  "phase": "B",
  "status": "pending_review",
  "source": "SRC-AGRI-GEN"
 },
 {
  "id": "PROC-WINDBREAK",
  "name": "Windbreak and ground cover",
  "applies_to": [
   "ARC-ARID",
   "ARC-SALINE",
   "ARC-SANDY"
  ],
  "targets": [
   "wind erosion",
   "wind damage"
  ],
  "prerequisites": [],
  "materials": [
   "tall hardy plants, brush or woven barriers"
  ],
  "rate": "1 barrier row per 10 m of exposed plot width, across the prevailing wind",
  "unit": "",
  "steps": [
   "Place barriers across the prevailing wind direction.",
   "Cover bare patches before planting."
  ],
  "verification": [
   "Visible dust movement reduced"
  ],
  "stabilization_days": 0,
  "phase": "A",
  "status": "pending_review",
  "source": "SRC-CASE"
 },
 {
  "id": "PROC-DRAINAGE",
  "name": "Drainage correction",
  "applies_to": [
   "ARC-WATERLOGGED"
  ],
  "targets": [
   "standing water",
   "slow drainage"
  ],
  "prerequisites": [],
  "materials": [
   "channels or drains, raised beds"
  ],
  "rate": "Channels about 30 cm wide and 30 cm deep, spaced 5-10 m apart, falling toward an outlet",
  "unit": "",
```

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 "steps": [
  "Dig channels from low patches toward an outlet.",
  "Do not plant until surface water is gone.",
  "Let soil dry until it crumbles rather than smears."
 ],
 "verification": [
  "No standing water",
  "Infiltration check",
  "Crumb test"
 ],
 "stabilization_days": 7,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-CASE"
},
{
 "id": "PROC-LEACH",
 "name": "Salinity leaching",
 "applies_to": [
  "ARC-SALINE"
 ],
 "targets": [
  "salinity"
 ],
 "prerequisites": [
  "Confirmed low-salt water source",
  "Confirmed drainage out of the root zone"
 ],
 "materials": [
  "low-salt water"
 ],
 "rate": 150,
 "unit": "mm of low-salt water per leaching event; repeat until crust is gone",
 "steps": [
  "Flush with low-salt water only if water and drainage are confirmed."
 ],
 "verification": [
  "EC measurement or crust gone"
 ],
 "stabilization_days": 7,
 "safety": [
  "Leaching without drainage raises the water table and brings salt back up."
 ],
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-AGRI-GEN"
},
{
 "id": "PROC-DECOMPACT",
 "name": "Decompaction and aeration",
 "applies_to": [
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "targets": [
  "compaction",
  "poor infiltration"
 ],
 "prerequisites": [
  "Soil not saturated"
 ],
 "materials": [
  "fork, broadfork or subsoiler"
 ],
 "rate": "Loosen to 25-30 cm depth",
 "unit": "",
 "steps": [
  "Loosen the soil when moderately moist, not wet.",
  "Follow immediately with organic matter."
 ],
 "verification": [
  "Infiltration check"
 ],
 "stabilization_days": 3,
 "phase": "A",
 "status": "pending_review",
 "source": "SRC-CASE"
},
{
 "id": "PROC-COVER-CROP",
 "name": "Cover crop / green manure",
 "applies_to": [
  "ARC-COMPACTED",
  "ARC-DISTURBED",
  "ARC-ARID",
  "ARC-SANDY"
 ],
 "targets": [
  "nutrient depletion",
  "bare soil",
  "organic matter"
 ],
 "prerequisites": [
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    "PROC-DECOMPACT or PROC-ORG-MATTER where relevant"
   ],
   "materials": [
    "legume cover crop seed"
   ],
   "rate": 5,
   "unit": "g legume seed per m2",
   "steps": [
    "Sow, grow, then cut and leave or dig in."
   ],
   "verification": [
    "Soil cover established"
   ],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-AGRI-GEN"
  },
  {
   "id": "PROC-SHADE",
   "name": "Temporary shade or surface protection",
   "applies_to": [
    "ARC-DISTURBED",
    "ARC-ARID"
   ],
   "targets": [
    "heat exposure"
   ],
   "prerequisites": [],
   "materials": [
    "shade cloth, mulch or nurse plants"
   ],
   "rate": "30-50% shade cloth or mulch cover over seedlings until established",
   "unit": "",
   "steps": [
    "Protect seedlings during establishment."
   ],
   "verification": [],
   "stabilization_days": 0,
   "phase": "B",
   "status": "pending_review",
   "source": "SRC-CASE"
  }
 ],
 "readiness_criteria": [
  {
   "id": "RC-CRUMB",
   "name": "Crumb structure test",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-COMPACTED",
    "ARC-DISTURBED",
    "ARC-WATERLOGGED"
   ],
   "check": "A handful of soil holds together and breaks into crumbs rather than falling apart as dust or smearing as
     paste.",
   "mandatory": true,
   "if_failed": "Add organic matter or let soil dry, then repeat."
  },
  {
   "id": "RC-INFIL",
   "name": "Infiltration check",
   "applies_to": [
    "ARC-WATERLOGGED",
    "ARC-COMPACTED",
    "ARC-DISTURBED"
   ],
   "check": "Water poured on the surface soaks in without pooling.",
   "mandatory": true,
   "if_failed": "Repeat drainage or decompaction."
  },
  {
   "id": "RC-NOPOOL",
   "name": "No standing water",
   "applies_to": [
    "ARC-WATERLOGGED"
   ],
   "check": "No standing puddles on the plot.",
   "mandatory": true,
   "if_failed": "Continue drainage correction."
  },
  {
   "id": "RC-MOIST",
   "name": "Moisture retention",
   "applies_to": [
    "ARC-SANDY",
    "ARC-ARID"
   ],
   "check": "Soil stays damp below the surface after watering.",
   "mandatory": true,
   "if_failed": "Add organic matter and mulch."
  },
```

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 {
  "id": "RC-DUST",
  "name": "Surface stability",
  "applies_to": [
   "ARC-ARID",
   "ARC-SANDY",
   "ARC-SALINE"
  ],
  "check": "Wind no longer lifts loose soil from the surface.",
  "mandatory": true,
  "if_failed": "Add windbreak or ground cover."
 },
 {
  "id": "RC-SALT",
  "name": "Salt crust cleared",
  "applies_to": [
   "ARC-SALINE"
  ],
  "check": "No visible crust and a measured EC reading is available.",
  "mandatory": true,
  "if_failed": "Provide an EC reading or continue salinity management."
 },
 {
  "id": "RC-STAB",
  "name": "Stabilization window",
  "applies_to": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-COMPACTED",
   "ARC-DISTURBED"
  ],
  "check": "Required waiting time after amendments has passed.",
  "mandatory": true,
  "window_days": 3,
  "if_failed": "Wait for the stabilization window to pass, then repeat."
 }
],
"crops": [
 {
  "id": "CROP-COWPEA",
  "name": "Cowpea / Black-eyed bean",
  "scientific": "Vigna unguiculata",
  "category": "Legume",
  "soil_compat": [
   "ARC-SANDY",
   "ARC-ARID",
   "ARC-DISTURBED"
  ],
  "tolerates": [
   "drought",
   "heat",
   "low fertility"
  ],
  "avoid": [
   "waterlogging",
   "salinity"
  ],
  "water": "low-medium",
  "light": "full",
  "duration_days": [
   60,
   75
  ],
  "feeder_class": "light",
  "role": "Pioneer crop; fixes nitrogen on poor land.",
  "planting": {
   "depth": "2-4 cm",
   "spacing": "10 cm in row, 40-50 cm between rows",
   "method": "Direct sow into moist, prepared soil"
  },
  "care": {
   "diseases": [
    "Fusarium wilt"
   ],
   "pests": [
    "Aphids",
    "Pod borers"
   ],
   "controls": [
    "Neem-based spray"
   ],
   "fertilizer": "Little nitrogen needed (fixes its own). Compost at bed prep; phosphorus if available."
  },
  "harvest": {
   "indicators": [
    "Pods dry and yellow-brown"
   ],
   "method": "Hand-pick pods as they dry"
  },
  "status": "pending_review",
  "source": "SRC-PRD",
  "phase_days": {
   "P1": "3-7",
```

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  "P2": "20-25",
  "P3": "20-30",
  "P4": "7-10"
 }
},
{
 "id": "CROP-SORGHUM",
 "name": "Sorghum",
 "scientific": "Sorghum bicolor",
 "category": "Cereal / dryland grain",
 "soil_compat": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "heat",
  "marginal soil",
  "some waterlogging",
  "moderate salinity"
 ],
 "avoid": [],
 "water": "low",
 "light": "full",
 "duration_days": [
  90,
  120
 ],
 "feeder_class": "moderate",
 "role": "Drought and marginal-soil tolerant staple grain.",
 "planting": {
  "depth": "2-4 cm",
  "spacing": "15-20 cm in row, 60-75 cm between rows",
  "method": "Direct sow into prepared soil"
 },
 "care": {
  "diseases": [
   "Grain mold",
   "Anthracnose"
  ],
  "pests": [
   "Shoot fly",
   "Stem borer"
  ],
  "controls": [
   "Neem-based spray",
   "Intercrop with cowpea to disrupt stem borer",
   "Seed treatment with Trichoderma",
   "Reflective ribbon against birds"
  ],
  "fertilizer": "Modest nitrogen and phosphorus; well-rotted compost at bed prep; light top-dress at knee-height
    stage."
 },
 "harvest": {
  "indicators": [
   "Grain hard, heads mature"
  ],
  "method": "Cut heads, dry before storage"
 },
 "status": "pending_review",
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
  "P3": "30-40",
  "P4": "10-14"
 }
},
{
 "id": "CROP-PMILLET",
 "name": "Pearl millet",
 "scientific": "Pennisetum glaucum",
 "category": "Cereal / dryland grain",
 "soil_compat": [
  "ARC-SANDY",
  "ARC-ARID",
  "ARC-SALINE",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "heat",
  "low fertility",
  "some salinity"
 ],
 "avoid": [
  "waterlogging"
 ],
 "water": "low",
 "light": "full",
 "duration_days": [
```

```
    65,
    90
   ],
   "feeder_class": "light",
   "role": "Drought and low-fertility tolerant cereal.",
   "planting": {
    "depth": "2-3 cm",
    "spacing": "10-15 cm in row, 45-50 cm between rows",
    "method": "Direct sow into prepared soil"
   },
   "care": {
    "diseases": [
     "Downy mildew",
     "Ergot"
    ],
    "pests": [
     "Stem borer"
    ],
    "controls": [
     "Clean or resistant seed",
     "Remove infected plants early",
     "Neem spray"
    ],
    "fertilizer": "Low demand; compost at bed prep and a small nitrogen top-dress at tillering."
   },
   "harvest": {
    "indicators": [
     "Ear heads dry, grains hard"
    ],
    "method": "Cut ear heads and dry"
   },
   "status": "pending_review",
   "source": "SRC-PRD",
   "phase_days": {
    "P1": "3-5",
    "P2": "20-25",
    "P3": "25-35",
    "P4": "7-10"
   }
  },
  {
   "id": "CROP-AMARANTH",
   "name": "Amaranth",
   "scientific": "Amaranthus spp.",
   "category": "Grain / leafy dual-purpose",
   "soil_compat": [
    "ARC-SANDY",
    "ARC-ARID",
    "ARC-DISTURBED"
   ],
   "tolerates": [
    "drought",
    "heat"
   ],
   "avoid": [],
   "water": "low-medium",
   "light": "full",
   "duration_days": [
    30,
    40
   ],
   "duration_note": "Leafy harvest 30-40 days; grain harvest 90-110 days.",
   "feeder_class": "light",
   "role": "Pioneer crop where soil test results are uncertain.",
   "planting": {
    "depth": "0.5-1 cm (tiny seed; mix with sand)",
    "spacing": "10-15 cm in row, 30-45 cm between rows; thin seedlings",
    "method": "Broadcast or drill shallowly into fine, moist soil"
   },
   "care": {
    "diseases": [
     "Damping-off",
     "Leaf spot",
     "Root rot in wet soil"
    ],
    "pests": [
     "Aphids"
    ],
    "controls": [
     "Neem spray",
     "Avoid overwatering",
     "Thin plants for airflow"
    ],
    "fertilizer": "Light feeder; compost at bed prep and a light nitrogen feed after the first leaf cut."
   },
   "harvest": {
    "indicators": [
     "Leaves tender (leafy)",
     "Seed heads dry (grain)"
    ],
    "method": "Cut leaves early or seed heads late"
   },
   "status": "pending_review",
```

```
 "source": "SRC-PRD",
 "phase_days": {
  "P1": "3-5",
  "P2": "15-25",
  "P3": "40-60 (grain)",
  "P4": "7-10"
 }
},
{
 "id": "CROP-RICE",
 "name": "Rice (paddy)",
 "scientific": "Oryza sativa",
 "category": "Cereal / wetland",
 "soil_compat": [
  "ARC-WATERLOGGED"
 ],
 "tolerates": [
  "waterlogging"
 ],
 "avoid": [
  "drought"
 ],
 "water": "high",
 "light": "full",
 "duration_days": [
  100,
  150
 ],
 "feeder_class": "moderate",
 "role": "Wet-footed cereal for waterlogged land that cannot be fully drained.",
 "planting": {
  "depth": "2-3 cm; transplant 2-3 seedlings per hill",
  "spacing": "20 x 20 cm",
  "method": "Transplant 20-30 day seedlings into puddled soil with shallow standing water"
 },
 "care": {
  "diseases": [
   "Blast",
   "Sheath blight",
   "Bacterial leaf blight"
  ],
  "pests": [
   "Stem borer",
   "Brown planthopper"
  ],
  "controls": [
   "Resistant varieties",
   "Balanced nitrogen",
   "Brief field drainage to disrupt pests",
   "Neem spray"
  ],
  "fertilizer": "Compost at bed prep; split nitrogen at tillering and panicle initiation."
 },
 "harvest": {
  "indicators": [
   "80-85% of grains straw coloured",
   "Grains hard"
  ],
  "method": "Drain field 7-10 days before; cut, dry, thresh"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
  "P3": "35-45",
  "P4": "10-15"
 }
},
{
 "id": "CROP-TARO",
 "name": "Taro",
 "scientific": "Colocasia esculenta",
 "category": "Root / tuber",
 "soil_compat": [
  "ARC-WATERLOGGED"
 ],
 "tolerates": [
  "waterlogging"
 ],
 "avoid": [
  "drought"
 ],
 "water": "high",
 "light": "full to partial",
 "duration_days": [
  180,
  240
 ],
 "feeder_class": "moderate",
 "role": "Wet-footed root crop.",
 "planting": {
  "depth": "5-10 cm",
```

```
  "spacing": "60 x 60 cm",
  "method": "Plant corm pieces or suckers on wet ridges"
 },
 "care": {
  "diseases": [
   "Taro leaf blight",
   "Corm rot"
  ],
  "pests": [
   "Aphids",
   "Armyworm"
  ],
  "controls": [
   "Remove infected leaves",
   "Use disease-free planting material",
   "Keep wide spacing"
  ],
  "fertilizer": "Compost-heavy; potassium-rich organic matter."
 },
 "harvest": {
  "indicators": [
   "Leaves yellow and die back",
   "Corms firm"
  ],
  "method": "Dig corms carefully; cure before storage"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "14-21",
  "P2": "60-90",
  "P3": "60-90",
  "P4": "14-21"
 }
},
{
 "id": "CROP-BARLEY",
 "name": "Barley",
 "scientific": "Hordeum vulgare",
 "category": "Cereal",
 "soil_compat": [
  "ARC-SALINE",
  "ARC-SANDY"
 ],
 "tolerates": [
  "salinity",
  "drought"
 ],
 "avoid": [
  "waterlogging"
 ],
 "water": "low-medium",
 "light": "full",
 "duration_days": [
  90,
  120
 ],
 "feeder_class": "moderate",
 "role": "Salt-tolerant cereal.",
 "planting": {
  "depth": "3-5 cm",
  "spacing": "Drill in rows 20 cm apart",
  "method": "Drill or broadcast and cover"
 },
 "care": {
  "diseases": [
   "Powdery mildew",
   "Net blotch",
   "Rust"
  ],
  "pests": [
   "Aphids",
   "Armyworm"
  ],
  "controls": [
   "Resistant varieties",
   "Crop rotation",
   "Neem spray"
  ],
  "fertilizer": "Moderate nitrogen, split; compost at bed prep; avoid chloride-heavy fertilizers."
 },
 "harvest": {
  "indicators": [
   "Heads golden",
   "Grains hard"
  ],
  "method": "Cut when dry, thresh, store below 12% moisture"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "5-10",
  "P2": "30-40",
```

```
  "P3": "30-40",
  "P4": "10-14"
 }
},
{
 "id": "CROP-PIGEONPEA",
 "name": "Pigeon pea",
 "scientific": "Cajanus cajan",
 "category": "Legume",
 "soil_compat": [
  "ARC-COMPACTED",
  "ARC-DISTURBED"
 ],
 "tolerates": [
  "drought",
  "compaction (deep roots)"
 ],
 "avoid": [
  "waterlogging"
 ],
 "water": "low",
 "light": "full",
 "duration_days": [
  150,
  180
 ],
 "feeder_class": "light",
 "role": "Deep-rooted legume that helps recover compacted, depleted soil.",
 "planting": {
  "depth": "3-5 cm",
  "spacing": "30 cm in row, 75-90 cm between rows",
  "method": "Direct sow into loosened soil"
 },
 "care": {
  "diseases": [
   "Fusarium wilt",
   "Sterility mosaic",
   "Phytophthora blight"
  ],
  "pests": [
   "Pod borer",
   "Pod fly",
   "Aphids"
  ],
  "controls": [
   "Neem spray",
   "Intercrop with cereals",
   "Resistant varieties",
   "Pheromone traps"
  ],
  "fertilizer": "Little nitrogen; phosphorus helpful; compost at bed prep."
 },
 "harvest": {
  "indicators": [
   "Pods brown and dry",
   "Seeds hard"
  ],
  "method": "Hand-pick or cut dry pods in passes; thresh"
 },
 "status": "pending_review",
 "source": "SRC-AGRI-GEN",
 "phase_days": {
  "P1": "5-10",
  "P2": "40-60",
  "P3": "50-70",
  "P4": "10-15"
 }
},
{
 "id": "CROP-TOMATO",
 "name": "Tomato",
 "scientific": "Solanum lycopersicum",
 "category": "Vegetable",
 "soil_compat": [],
 "tolerates": [],
 "avoid": [
  "drought",
  "low fertility",
  "salinity"
 ],
 "water": "high",
 "light": "full",
 "duration_days": [
  90,
  120
 ],
 "feeder_class": "heavy",
 "role": "Demonstrates the heavy-feeder safety filter. Excluded on depleted soil.",
 "planting": {
  "depth": "0.5-1 cm in nursery; transplant at 5 cm",
  "spacing": "45 x 60 cm",
  "method": "Raise seedlings, then transplant"
 },
```

```
  "care": {
   "diseases": [
    "Early blight",
    "Late blight",
    "Bacterial wilt"
   ],
   "pests": [
    "Whitefly",
    "Fruit borer"
   ],
   "controls": [
    "Neem spray",
    "Crop rotation",
    "Stake and prune for airflow"
   ],
   "fertilizer": "Heavy feeder: rich compost plus balanced NPK, split through the season."
  },
  "harvest": {
   "indicators": [
    "Fruit fully coloured and slightly firm"
   ],
   "method": "Hand-pick ripe fruit regularly"
  },
  "status": "pending_review",
  "source": "SRC-AGRI-GEN",
  "phase_days": {
   "P1": "5-8",
   "P2": "25-30",
   "P3": "30-45",
   "P4": "7-14"
  }
 }
],
"phases": {
 "journey": [
  {
   "id": "P1",
   "name": "Germination"
  },
  {
   "id": "P2",
   "name": "Vegetative Growth"
  },
  {
   "id": "P3",
   "name": "Reproductive & Canopy Development"
  },
  {
   "id": "P4",
   "name": "Pre-Harvest"
  }
 ],
 "note": "Crop-specific durations come from each crop record. If duration_days is null, the Timeframe step must say the
    data is unavailable."
},
"missing_data_messages": {
 "no_rate": "The Knowledge Base has no verified application rate for this treatment. No quantity will be suggested.",
 "no_crop_duration": "The Knowledge Base has no verified growing duration for this crop.",
 "no_planting_spec": "The Knowledge Base has no verified planting depth or spacing for this crop.",
 "no_readiness_window": "The Knowledge Base has no verified stabilization window for this condition.",
 "low_evidence": "Not enough evidence to classify this land. Please add a description, image or measurement."
},
"scenarios": [
 {
  "id": "SC-1",
  "name": "Arid Wasteland",
  "expect_archetypes": [
   "ARC-ARID",
   "ARC-SANDY"
  ],
  "expect_features": [
   "F-CRACKED",
   "F-LOOSE-DUST",
   "F-NO-VEG",
   "F-HIGH-HEAT",
   "F-RAIN-RARE"
  ]
 },
 {
  "id": "SC-2",
  "name": "Flood-Damaged Wetland",
  "expect_archetypes": [
   "ARC-WATERLOGGED"
  ],
  "expect_features": [
   "F-DARK-STICKY",
   "F-STANDING-WATER",
   "F-SOUR-SMELL",
   "F-YELLOW-WEEDS",
   "F-SLOW-DRAIN"
  ]
 },
 {
```

```
    "id": "SC-3",
    "name": "Coastal Saline Land",
    "expect_archetypes": [
     "ARC-SALINE",
     "ARC-SANDY"
    ],
    "expect_features": [
     "F-GRITTY",
     "F-SALT-CRUST",
     "F-SALT-SMELL",
     "F-STUNTED-BROWN",
     "F-SEA-NEAR"
    ]
   },
   {
    "id": "SC-4",
    "name": "Overgrazed Pastureland",
    "expect_archetypes": [
     "ARC-COMPACTED"
    ],
    "expect_features": [
     "F-HARD-PACKED",
     "F-SURFACE-POOLING",
     "F-BARE-PATCHES",
     "F-THIN-GRASS"
    ]
   },
   {
    "id": "SC-5",
    "name": "Post-Deforestation Cleared Land",
    "expect_archetypes": [
     "ARC-DISTURBED"
    ],
    "expect_features": [
     "F-STUMPS",
     "F-NO-CANOPY",
     "F-HARD-PACKED"
    ]
   }
  ]
}
```